SPACE EMANATION THEORY

Source Energy, Electromagnetic Evolution and Magnetic Organization

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From capacity and complete source accounting to signed currents, material work and a calculated magnetic allocation.

The same energy that generates clocks and material motion must account for the field those sources establish. This research follows that requirement into the electromagnetic sector, tests the planetary rotational-clock candidate, and derives organized-field energies in explicitly supplied material states.

Principal result
A finite charged-material preparation can calculate its magnetic share through reciprocal evolution. The field geometry and driver recoil determine the allocation. A separate positive variational calculation determines the least energy needed to sustain a specified dipole.

The development retains source identities, conditional field selections, numerical checks and empirical comparisons in one sequence. Every additional material input and approximation is stated. The work continues the published *Source, Clock and Interacting-Field Energy* synthesis.

Contents

1. The question and the retained physical reading 3
2. The retained source and the electromagnetic normalization 4
3. What common evolution selects about charged coupling 5
4. Signed internal state, neutral evolution and the charge still supplied 7
5. An explicit source functional that generates charged evolution 8
6. Positive source shares, material clocks and field stress 10
7. Dynamical selection: energy, momentum and propagation 11
8. Field memory, null pulses and source-to-photon work 13
9. Ordered quantum motion and changing clock-energy shares 14
10. Signed material motion is the source of a magnetic moment 16
11. Burden gradients supply calculable material work 17
12. Thermal, phase and magnetic energy must share one source ledger 18
13. What the material and induction calculations establish 20
14. The rotational clock expression and its magnetic reading 21
15. What the unchanged planetary and solar calculation gives 22
16. Measurement history and physical source history 23
17. Testing source organization without a fitted efficiency 25
18. Projecting the common energy onto magnetic modes 26
19. The energy cost of an exterior magnetic harmonic 27
20. An exact constrained organization of a neutral current 28
21. The complete source retains physical initial-state freedom 30
22. A regular current mode with a calculated magnetic share 31
23. Exact inductive geometry for a finite material class 33
24. A finite driver calculates the allocation 34
25. Calculated organization, convergence and limits 36
26. An additional capacity cross-term and its numerical test 38
27. Reconciling the phase connection, clock weighting and charged exterior 39
28. What determines the organized magnetic energy 41
29. Verification, retention and reproducibility 42
30. References and reproducibility 42

1. The question and the retained physical reading

Mass continuously commits field interaction capacity as gravitational burden. Motion and the local progression of physical events share the remaining capacity. SET expresses that relation through the invariant and measures a specified source through Q. The present question is what these same tools determine when part of the source is electrically charged, moves internally and establishes a magnetic field.

The task is to calculate how much complete source energy becomes an organized field. Total energy, magnetic storage and the exterior dipole are different quantities. A source can retain its complete energy while current changes, heat is produced or magnetic energy moves between interior structure and the field measured outside it. The calculation must follow those transfers.

This paper retains the research completed after *Source, Clock and Interacting-Field Energy* [R1], then develops the magnetic allocation question. Each calculation states its field and material variables, its approximation and the input data. Algebraic identities are carried forward exactly. Added test geometries and measured material properties are identified where they enter. No magnetic amplitude or efficiency is fitted to an observed target.

The results to carry forward

A complete source functional generates charged evolution. The charged spherical functional includes the field tail, source-dependent integrating factor and material phase. Its full variation supplies the same clock, work and transport response.

Source, clock and momentum tests do different work. Supported boundary comparisons constrain electromagnetic normalization inside the retained class. Electric energy exchange fixes reciprocal coupling; momentum exchange also tests magnetic turning, which scalar work alone cannot select.

Material organization gives a calculable magnetic partition. A regular neutral two-species mode yields an exterior fraction 7/30 of its magnetic energy and a separate inductive-to-material allocation. A finite driver supplies that energy through reciprocal evolution.

The least energy required for a dipole can be solved exactly. The admitted positive current class has a unique constrained minimizer. Its modified spherical Bessel profile yields a sharp, material-dependent bound on the exterior field without fitting a magnetic efficiency.

Identical total source readings can support different magnetic states. An explicit full-constraint initial-data family and two finite-driver preparations demonstrate how signed state and history enter. The common energy selects their evolution once that state is supplied.

The planetary numbers sharpen the problem. Earth’s close rotational-clock scale is retained, as are the solar series and the large failures elsewhere. Measurement history, independent source structure and complete field energy separate these effects.

Quantum clocks and photon propagation remain tied to the generator. Ordered positive energy identities survive noncommutation. Actual evolving-source ray calculations retain endpoint clocks and changing propagation, providing checks beyond a scalar exposure argument.

How to read the notation

Symbol	Meaning in this paper
S; β; α or N	Burden in speed units; material clock rate dτ/dT; held-clock lapse. In a zero-shift supplied reading, $S^2=c^2(1-N^2)$.
Q_s; Q_0; Q_m	Complete source rate; fixed rest-source label of a massive probe; its moving-clock reading Q_0/β. Electric charge is written e, q or q_e and is distinct.
V_ref; H_complete	Declared source reference volume; complete generator including material, support, field, binding and exported quantities when applicable.
q_{ij}; a; R	Spatial ruler; a radial ruler or a specified source radius according to its section; an independently stated comparison radius.
B; b; U_B; U_out	Magnetic vector field; equatorial dipole amplitude; complete magnetic energy; field energy exterior to a stated sphere.
κ; k; m or m	Carrier-motion coefficient; inverse material length; magnetic dipole moment in the current sections. Material mass is identified separately.

SI units are used for observations and the explicitly dimensional current examples. The charged spherical functional and selected quantum controls declare their normalized units. Several familiar symbols are reused in their standard local roles; every such role is defined before the derivation. The supplementary record preserves the complete algebra, numerical inputs and verification scripts.

2. The retained source and the electromagnetic normalization

The published source-and-clock synthesis [R1] established a common-energy construction within a stated local field and material realization. The present work begins from that result. Its two scalar readings are capacity and complete source energy. The first compares physical motion with internal clock progression; the second assigns the complete energy of a specified source to its source-rate quotient. Electric charge is a different quantity.

$$\begin{aligned} S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 &= c^2, \\ \frac{Q_s^2}{V_{\mathrm{ref}}} &= \frac{24\pi G}{c^2} H_{\mathrm{complete}}. \end{aligned} \tag{1}$$

Here Q_s is the source rate, V_ref is its declared reference volume, and H_complete includes material, supports, field energy and binding. In the retained canonical description one energy supplies velocity H_p and clock rate H_E0. Those variables and brackets are part of the realization. They are not obtained merely by multiplying the scalar capacity equality by a dimensional factor. The later research asks what this construction selects about electromagnetism, charged matter and actual field organization.

The first distinction is between light speed and field calibration [N1]. The scalar speed fixes the product of permittivity and permeability. It does not fix their ratio. In the admitted positive quadratic vector class, common generator composition gives $ad=c^2$ and therefore the following energy with one remaining impedance Z:

$$c^2 = \frac{1}{\varepsilon\mu}, \qquad Z = \sqrt{\frac{\mu}{\varepsilon}}, \\ u = \frac{aD^2 + dB^2}{2} = \frac{c}{2}\left(ZD^2 + \frac{B^2}{Z}\right). \tag{2}$$

The independent supported-source test in [R1,N1] adds information that a redefinition of H cannot. Field equations and material supports are solved first; their boundary source reading is then compared with the energy reading. For the tested weak neutral-source family, the resulting local restrictions make Z constant on the connected domain reached by that family. They do not derive its numerical SI value or exclude every field theory outside that class. Coordinate optical coefficients can still vary together with lapse and ruler: their ratio remains Z while coordinate light speed changes.

A constant field rescaling makes the remaining calibration freedom explicit. With $A'=sA$, $D'=D/s$ and $B'=sB$, the same energy and interaction have $Z'=s^2Z$ and $e'=e/s$. The dimensionless combination $e^2Z/(4\pi\hbar)$ is unchanged. The later charged calculation must therefore determine a physical interaction strength, if it is to do so at all; renormalizing the field variables cannot accomplish it.

There is also an exact electromagnetic scalar identity. Writing the Poynting vector as P, direct expansion gives

$$u^2 - \frac{|\mathbf{P}|^2}{c^2} = \frac{(E^2 - c^2B^2)^2 + 4c^2(\mathbf{E}\cdot\mathbf{B})^2}{4Z^2c^2} \geq 0. \tag{3}$$

This permits a capacity-shaped decomposition into energy flux and a positive remainder. It does not make that remainder a material clock or select Maxwell evolution. A uniform electric vector can rotate in time while preserving this scalar equality and violating vacuum Ampère evolution. Likewise, normalizing the gravitational work field into electric units matches a quadratic expression but does not turn attractive source response into electric charge. Field identities, field equations and material interpretation have to be tested separately.

3. What common evolution selects about charged coupling

Admit positive real canonical matter φ,π and the already retained Maxwell vector field. In the regular first-class branch, allow independent coefficients for the terms linear and quadratic in the vector potential [N2]. For a two-component internal plane, J rotates the components through a right angle. Before imposing a coupling square, the gradient energy is

$$h_{\mathrm{grad}} = \frac{1}{2}\sum_i \left[|\partial_i\phi|^2 + 2bA_i(J\phi)\cdot\partial_i\phi + dA_i^2|\phi|^2\right], \\ J(\phi_1,\phi_2) = (-\phi_2,\phi_1). \tag{4}$$

The common-generator bracket identifies the Gauss constraint. Requiring it to remain preserved for generic matter data makes its generated internal transformation preserve the energy. The coefficients of independent local variations then give b=−e and d=e². Thus the linear and quadratic terms are related rather than independently adjustable:

$$h_{\mathrm{grad}} = \tfrac{1}{2}\sum_i |\partial_i \phi - eA_i J\phi|^2,$$
$$T^T = -T, \qquad C = T^T T, \qquad (T\phi)\cdot\nabla_\phi U = 0. \tag{5}$$

The second line is the corresponding matrix result. A homogeneous internal rotation is selected within this matter class, together with its quadratic partner and an invariant potential. A single real component has no nonzero homogeneous antisymmetric generator. Extra fields, affine generators, different constraint branches and higher-derivative or nonminimal interactions require separate analysis. This result connects standard minimal-coupling structure to the retained common evolution; it does not derive the admitted internal plane from Q_s.

Capacity adds a different restriction. Consider a positive probe Hamiltonian with an energy-dependent momentum shift a and additive term u. Its velocity already satisfies capacity with β_0. The actual derivative clock differs by Δ, giving an exact residual:

$$H = \alpha\sqrt{E_0^2 + c^2|p - a(E_0)|^2} + u(E_0),$$
$$\Delta = u_{E_0} - V^i a_{i,E_0}, \qquad \mathcal{R}_{\mathrm{cap}} = c^2\Delta(2\beta_0 + \Delta). \tag{6}$$

The second zero of this residual reverses the clock sign. On the positive branch, zero kinetic momentum p−a=0 gives u_E0=0 and variation over the open velocity ball then gives a_E0=0. A fixed electric coupling survives, including zero. This derivative varies one clock-energy label at fixed charge sector; it is not a universal relation between the masses and charges of different species. Identifying a raw internal phase rate with the clock would add a different operational assumption.

Once the coupling is admitted, one Hamiltonian supplies field charge, transport current and material response. Electric work requires the same coefficient in source and force. The charged Euler identity retains the term that the neutral formula would omit:

$$H = E_0\beta + p_i V^i + eH_e = \frac{Q_0}{2}H_{Q_0} + p_i V^i + eH_e,$$
$$H_e = \Phi - A_i V^i, \qquad \frac{d(\alpha^2 Q_m)}{dT} = \frac{eQ_0}{E_0}E_{Ti}V^i. \tag{7}$$

The work law assumes fixed E_0,Q_0 and a static zero-shift background. In an electrostatic field its conserved expression includes $(eQ_0/E_0)\Phi$. In a purely magnetic field the scalar energy reading stays fixed while momentum turns. Evolving geometry adds explicit lapse and ruler work. These identities reconcile charge transport with source accounting without selecting the magnitude of e.

4. Signed internal state, neutral evolution and the charge still supplied

Positive energy and signed charge need not have the same algebraic character. For a degenerate two-dimensional harmonic internal system, completing positive squares yields two nonnegative actions [N3,N4]:

$$E_{\mathrm{int}} = \omega(I_+ + I_-), \qquad L = I_+ - I_-, \\ I_\pm \geq 0, \qquad |L| \leq E_{\mathrm{int}}/\omega. \tag{8}$$

At fixed energy the whole allowed signed interval occurs. A source reading can therefore count an energy sum while a signed interaction reads an imbalance. The electromagnetic assignment q_e=(e_SI/ħ)L is supplied, however; it is not obtained from the sum. Unequal internal frequencies preserve positive energy but give L=(ω₁²−ω₂²)xy. Capacity and positive source energy alone do not force the degeneracy.

The later internal-symmetry audit sharpens this statement [N5]. If observations require a nonzero coupling in the retained regular Gauss branch, constraint propagation requires the potential to preserve that internal rotation. A quadratic doublet must then be degenerate. This is a conditional selection after the charged branch is admitted. Explicit neutral supported sources with unequal internal frequencies still satisfy the independent source-boundary test. Their survival shows why the source law alone does not supply the missing representation. It also prevents confusing internal representation degeneracy with equality of all atomic transition energies.

The actual charge in this branch is an electric boundary flux. Its Gauss generator needs the boundary term that makes its variation well defined; continuity then includes the same lapse and ruler factors as matter evolution. No additional source-rate multiplier belongs in that charge. At fixed reference volume, replacing the energy generator by its positive square-root source variable gives X_Q=(Q/2H)X_H. This reparameterization has the same symmetry stabilizer as H. It cannot supply an internal rotation that the material energy lacks. The separate, independently solved source-boundary comparison retains its discriminating role because it is not this reparameterization.

Polar canonical variables make the actual signed transport visible. With φ=r(cosθ,sinθ), the admitted matter energy and current read

$$h = \frac{p_r^2}{2} + \frac{p_\theta^2}{2r^2} + \frac{|\nabla r|^2}{2} + \frac{r^2}{2}|\nabla\theta - e\mathbf{A}|^2 + U(r), \\ \rho_e = -e p_\theta, \qquad \mathbf{j} = e r^2(\nabla\theta - e\mathbf{A}), \\ \dot{\theta} + e\Phi = p_\theta/r^2. \tag{9}$$

Hamilton evolution gives charge continuity. A phase-gradient initial state develops oppositely signed local regions while total charge remains zero. Its refined lattice control retained Gauss residual below 1.5×10^{-14} and reduced energy error to 2.66×10^{-7}. The initial state already contained a phase current and a charged representation. It demonstrates charge

rearrangement, not creation of the representation from unsigned capacity. An exactly neutral invariant sector evolves with its second real component and electromagnetic fields zero.

Two further controls locate the freedom. At fixed phase action, the phase energy has a positive constrained minimum; it does not establish a dynamical attractor. In complete neutral capacitor sources, fixed material populations and an unchanged support law admit an open interval of different couplings, with different equilibrium radii and source energies. All pass the independent boundary equality. Changing electric loading changes Q_s through the same complete energy, so demanding unchanged Q_s would add a new constraint. Minimizing over a fundamental coupling with no dynamical equation would also be a new principle.

Observed Aharonov-Bohm and Josephson responses constrain the same spatial and temporal phase connection. Charged spin-zero particles prevent identifying that internal rotation with ordinary spatial spin. A compact phase can restrict relative representation weights without fixing its overall charge scale. A supplied charged condensate has a transverse gap proportional to its coupling times amplitude; a common slower clock does not remove that gap. Conversely, a smooth connection built from at most three scalar state labels obeys F∧F=0 and cannot represent generic electric and magnetic fields. These are bounded tests of proposed constructions, not a theorem against every possible completion.

The hydrogen and pair-threshold controls have the same status. Standard quantization of the admitted Coulomb branch gives 13.5982872641 eV for the leading hydrogen ionization energy, about 0.001083% below the cited value. Measured masses, charge and $\hbar$ enter; the agreement is a consistency check of that supplied branch, not an independent SET precision prediction. Charge remains conserved in ionization, and an allowed pair-production threshold does not determine its amplitude.

5. An explicit source functional that generates charged evolution

The dynamical research advances beyond stationary compatibility by giving the complete source quotient as a functional of an actual charged state [N6]. In this section G=c=1. Use a spherical real doublet φ,Π, canonical measure 4πr²dr and J as above. To avoid confusing a radial integrand with magnetic field, denote it by B_cal. Define

$$\mathcal{A} = |\Pi|^2 + |\phi_r|^2, \qquad q(r) = -4\pi g \int_0^r s^2 \Pi \cdot J\phi \, ds,$$
$$K(r) = \exp\left[-4\pi \int_r^\infty s\mathcal{A}(s)\, ds\right], \tag{10}$$
$$\mathcal{B}(r) = 2\pi r^2(\mathcal{A} + \mu^2|\phi|^2) + \frac{q^2}{2r^2}.$$

The radial constraints determine the mass and lapse from these fields. With a regular center and the complete exterior included, the result is

$$\begin{aligned} \mathcal{Q} \equiv \frac{Q_s^2}{V_{\mathrm{ref}}} &= 24\pi \int_0^\infty K\mathcal{B}\, dr, \\ (Km)_r = K\mathcal{B}, \quad m(0) &= 0, \quad f = 1 - 2m/r, \\ a = f^{-1/2}, \qquad N &= K\sqrt{f}. \end{aligned} \tag{11}$$

The state dependence of K is essential. Varying the entire functional, including both the integrating factor and the enclosed-charge integral, gives the coupled equations. The K variation produces the factor f in the kinetic and gradient response:

$$\begin{aligned} \phi_T &= Kf\Pi - g\Phi J\phi, \\ \Pi_T &= r^{-2}\partial_r(r^2 Kf\phi_r) - K\mu^2\phi - g\Phi J\Pi, \\ \Phi(r) = \int_r^\infty \frac{Kq}{s^2}\, ds, \qquad \phi_T &= \frac{1}{96\pi^2 r^2}\frac{\delta \mathcal{Q}}{\delta \Pi}. \end{aligned} \tag{12}$$

The negative φ derivative supplies Π_T. Freezing K while varying would discard the strong gravitational response. A conserved numerical value of Q_s²/V_ref does not mean that its functional derivative vanishes: many states on one energy surface have different velocities generated by the same functional. The admitted scalar representation, canonical bracket, positive quadratic mass term, charge calibration and boundary conditions remain explicit inputs. A noncompact scalar source has no exact finite material surface, so its V_ref is a declared reference volume.

The same expression provides a particularly transparent coupling audit. Hold the canonical profiles and phase normalization fixed, write q=gq_u, and separate the electric contribution:

$$\begin{aligned} H(g) &= H_0 + g^2 C, \qquad C = \int_0^\infty \frac{Kq_u^2}{2r^2}\, dr \geq 0, \\ m(r;g) &= m_0(r) + g^2 m_2(r), \qquad m_2 \geq 0. \end{aligned} \tag{13}$$

Constraint data regular at one g remain regular for smaller |g|, including zero. Independent reconstructions verify the quadratic dependence and the exterior tail. This is a family of consistent initial states, not a claim that each is the same equilibrium or remains regular forever. Multiple charged components produce a positive semidefinite quadratic form whose cross terms can have either sign. Net neutrality alone leaves internal electric energy; it vanishes only when the entire enclosed-charge profile cancels.

Regularity also imposes a real restriction. Where m_2 is positive, g^2 must stay below $(r-2m_0)/(2m_2)$ at every radius. The smallest allowed value is profile dependent. Approaching an attained threshold makes f and N tend to zero. This is a bound on admitted source data, not a universal maximum electric charge or a dynamical collapse theorem.

6. Positive source shares, material clocks and field stress

A common-energy interpretation requires an operational rest-energy label. Calibrate μ in unloaded test matter while holding other couplings fixed. The clock derivative then gives ε_μ=n, so the material energy density has the form ε=μn+e(n,σ) [N6]. This identifies a rest-energy channel; it does not select the residual equation of state e. Before that calibration, a clock condition fixes only a product of material and source factors, which cannot be split into two separately derived equalities.

$$\begin{aligned} \varepsilon = \mu n + e(n,\sigma), \qquad \Theta = \tfrac{e_\sigma}{n}, \\ \beta\Theta = \text{constant} \quad \text{at equilibrium.} \end{aligned} \tag{14}$$

Allowed equilibrium entropy exchange therefore gives a constant βΘ, or NΘ for held matter. This is the equilibrium clock weighting of heat, not a heat-transport law. In the stated positive regular fluid class, the full energy can also be separated exactly:

$$\begin{aligned} M &= \sum_i \int \mu_i \beta_i \, dN_i + \sum_i \int \Pi_i \tfrac{N v_i}{a} \, dN_i \\ &\quad + U_{\mathrm{EM,\,ref}} + \int N a \, e \, dV_c - G\partial_G M, \\ -G\partial_G M &= \int (1 - Na)\mathcal{E} \, dV_c \geq 0. \end{aligned} \tag{15}$$

The virtual G derivative holds canonical state fixed. Its positive binding sensitivity is global; it is not a unique local gravitational energy density. The momentum-velocity share approaches twice the ordinary quadratic kinetic energy. Positivity of the thermodynamic share uses e≥0. Subtracting σΘ does not generally preserve it, even for a convex equation of state. Independent reconstruction of evolved states checked eighteen derivatives to 1.3×10^{-12} and the summed shares to 4.1×10^{-15}; evolution accuracy requires separate refinement.

A radial field selector shows precisely what these clocks add. Admit electric energy u(q,r) independent of the radial ruler at fixed canonical charge, and compare arbitrary localized material variations after the intrinsic clock calibration. Matching their source weight requires Γ=Na. The lapse equation then gives P_r=−u, while the conjugate force is $4\pi r^2\partial_q u$. For u=f(D)/(4π), D=q/r², this becomes E=f′(D) and P_t=[Df′−f]/(4π). Electric force and stress are related by one energy; nonlinear f still survives this test. The stronger propagation condition below supplies the additional Maxwell restriction.

A scalar internal phase gives another useful split. Define D=∫4πr²φ·Πdr and I_μ=μ∂_μM. Then

$$\begin{aligned} M &= I_\mu + \mathcal{P} + \tfrac{1}{2}\dot{D}, \\ \mathcal{P} &= U_{\mathrm{EM}} + \int 2\pi K r^2 \left[(1-f)|\Pi|^2 + (1+f)|\phi_r|^2\right] dr \geq 0. \end{aligned} \tag{16}$$

For a stationary state Ɗ vanishes and the phase-family first law is dM=ωdN_phase, with N_phase=q/g in that convention. During actual deformation Ɗ can have either sign. A stationary positive partition cannot be applied instantaneously after dropping this term. Its quantized interpretation would again require a supplied action scale and representation.

Stress is equally essential. A radial electric field has P_r=−u, so it cancels from the particular combination energy plus radial pressure while still contributing to mass. In a general evolving zero-shift geometry, local electromagnetic work includes material, lapse and ruler terms:

$$\begin{aligned}\partial_T(N\sqrt{q}\,u)+\partial_i(N^2\sqrt{q}\,P^i)\\ =-N^2\sqrt{q}\,\mathbf{j}\cdot\mathbf{E}+N_T\sqrt{q}\,u\\ -\tfrac{N\sqrt{q}}{2}T^{ij}_{\mathrm{EM}}q_{ij,T}.\end{aligned}\tag{17}$$

For regular spherical sources with nonnegative mass and energy plus radial pressure, the calibrated source equations give 0<N≤1 and β=N/W≤1. This is a statement about the regular branch. It does not guarantee that arbitrary evolution remains there. Nor does a freely chosen synchronous coordinate lapse of one imply absent gravity: the spatial geometry can carry the response.

A charged exterior supplies another necessary bookkeeping correction. In SI its mass inside a screen is M∞−q_e²/(8πε₀c²r). The missing exterior energy contributes to the complete source quotient:

$$\begin{aligned}Q^2_{\mathrm{complete}}-Q^2_{\mathrm{screen}}=24\pi G\tfrac{E_{\mathrm{out}}}{c^2}V_{\mathrm{ref}},\\ 2\tfrac{dQ_s}{Q_s}-\tfrac{dV_{\mathrm{ref}}}{V_{\mathrm{ref}}}=\tfrac{dH_{\mathrm{complete}}}{H_{\mathrm{complete}}}.\end{aligned}\tag{18}$$

Here Q_screen denotes the finite-screen source reading, not the held-clock rate Q_s/N. Thus a finite charged screen cannot stand in for the complete asymptotic source without its tail. A changing reference volume also requires the quotient rather than Q_s alone. These distinctions will matter again when a nonspherical magnetic exterior is compared with a spherical source reading.

7. Dynamical selection: energy, momentum and propagation

Actual finite drivers test whether the terms above exchange energy and momentum consistently [N6]. If the field is sourced with one charge but material electric response is multiplied by λ, the radial energy balance contains an uncancelled term:

$$\tfrac{dM}{dT}=(1-\lambda)\int\tfrac{Na\,qq_T}{r^2}\,dr+\text{boundary terms.}\tag{19}$$

Generic electric work therefore selects λ=1 when no other exchange channel is omitted. Magnetic turning is subtler because its force is perpendicular to velocity. Runs with magnetic response factors zero, one and two all conserved their total model energy near 10^{-13} and satisfied the scalar capacity relation. The wrong factors nevertheless produced momentum drift about 2.88×10^{-6}, whereas the common-generator value conserved momentum near 10^{-17}. Energy conservation alone cannot select a perpendicular force.

Propagation provides a further independent selector. In c=1 units with compatible electric and magnetic normalization, take a smooth local abelian Lagrangian L(I,J), with $I=(E^2-B^2)/2$ and $J=E\cdot B$, no derivatives of the field tensor and no extra fields, requiring both physical polarizations to share the material null cone on every background forces the Hessian of L to vanish:

$$L = aI + \theta J + b_0. \tag{20}$$

The remaining constants include field normalization, a topological or boundary term and a vacuum offset. This is stronger than merely requiring a maximum speed c or no birefringence. Born-Infeld response is a positive clock-consistent countercontrol under the weaker requirements. A physical burden dependence of the electromagnetic coefficients adds terms to the source equation; keeping the same generic source law requires those coefficients to be constant. A mere variable transformation remains allowed when all induced terms are retained.

The dynamical controls include regular cold fluids, a thermal extension, finite charged scalar drivers and compact strong packets. Reversing a scalar driver changes its interaction and the final source response; any initial electric cross-energy difference is explicitly retained. The main accepted outcomes and numerical limits are kept together here so that algebraic precision is not confused with continuum accuracy.

Control	Retained result and numerical scope
Finite scalar drivers	At T=150, energy errors 3.79×10^{-12} and 3.51×10^{-12}; final primary amplitudes 0.0212286 and 0.0185301 from initial 0.02.
Compact strong packet	Minimum lapse 0.01119; maximum 2m/r=0.92758. Constraint-residual refinement factors 4.06 and 4.02, independently of stored-energy roundoff.
Cold regular material	Retain the resolved T=4 interval. Later T=5.5 compression with a nonconvergent roughly 1.3% energy deficit is rejected.
Thermal material	First-order, diffusive controls reduce energy error from 0.8475% to 0.1196% on 512-4096 cells. Global entropy increases; small local negative residuals diminish but do not establish a local entropy theorem.

These experiments verify a specified interacting completion. Their equations of state, charged representations and finite initial data are declared inputs. The result to carry forward is that the same complete source functional can generate and account for actual changes of state, while separate work, momentum and propagation tests distinguish requirements that a single scalar equality cannot.

8. Field memory, null pulses and source-to-photon work

Eliminating retained finite-profile Maxwell modes gives a causal self-response for an admitted weakly moving charged shell [N6]. Its delay is the light-crossing time, and no damping constant is fitted:

$$F_{\mathrm{self}} = K_d[v(T-\tau) - v(T)], \qquad \tau = 2a/c. \tag{21}$$

A positive finite-history storage contains the velocity squared integrated over this interval. Slow-history expansion yields electromagnetic inertia and a Schott-type term. An actual neutral contact driver loses the matching impulse and energy. Support stress is required in the complete accounting, including the familiar electromagnetic four-thirds issue. This effective history allocation is not literally the Maxwell energy on every simultaneous laboratory slice. The profiles remain declared rigid approximations; their exact clock derivatives do not solve relativistic material elasticity.

The transverse Maxwell Green tensor also yields the Darwin interaction and finite-profile self-inertia, checked by evolving modes with their moving positions retained. Magnetic force can be nonzero while its direct work vanishes; induction supplies the reciprocal energy transfer. This is the bridge from a field that stores energy to a field that remembers and mediates the actual motion of its sources.

For a photon through the same evolving spherical source, define C_γ=cN/a. Hamilton evolution gives an endpoint-clock factor and an integral over the actual changing geometry:

$$\frac{E_o}{E_e} = \frac{N_e}{N_o} \exp\left[\int_{\mathrm{ray}} \partial_T \ln C_\gamma \, dT\right]. \tag{22}$$

Neighboring pulse arrival intervals provide a separate clock test. In a refined intermediate ray, the energy ratio 0.759757190 agreed with the interval ratio 0.759758679, with second-order improvement as pulse separation was halved. Different screens in one evolved source gave ratios near 0.96723, 0.76466 and 1.07854. A stationary equal-lapse delay gave ratio one. Residence time alone therefore does not determine an attenuation factor. First-order reciprocal source work equals the negative reference-energy change of the photon; finite-beam backreaction and cosmological brightness are additional calculations.

An exact self-gravitating plane electromagnetic pulse supplies a complementary nonlinear control. It has energy and tidal focusing while its Kretschmann scalar is zero. A spherical calibration based on that scalar cannot be a universal detector of all field energy. Null-probe response depends on propagation direction, and after the pulse a relative velocity can persist in a flat region. This separates field memory from permanently retained local radiation energy. The result preserves the source-to-light calculation while ruling out an overgeneralized curvature or exposure shortcut.

9. Ordered quantum motion and changing clock-energy shares

The post-baseline quantum calculations test how the same source normalization survives noncommuting motion [N6]. First admit a spinless charged square-root Hamiltonian in a uniform static magnetic background and constant lapse N. The magnet and its support are external in this calculation. With kinetic momentum π and $E_0=mc^2$, define

$$\begin{gathered} H_T = NH_0, \qquad H_0 = \sqrt{E_0^2 + c^2\pi^2}, \\ \beta = NE_0/H_0, \qquad Q_m = Q_0/\beta, \qquad V_i = (i/\hbar)[H_T, x_i]. \end{gathered} \tag{23}$$

The exact anticommutator identity $\{H_0,V_i/N\}=2c^2\pi_i$ gives a symmetric source-momentum law. It also identifies the ordered object needed in the capacity expression:

$$\begin{gathered} \pi_i = \tfrac{m}{2Q_0}\{Q_m, V_i\}, \qquad U_i = \tfrac{1}{2}\{Q_m, V_i\}Q_m^{-1}, \\ S^2 I + \sum_i U_i^\dagger U_i + c^2 Q_0^2 Q_m^{-2} = c^2 I. \end{gathered} \tag{24}$$

Multiplication back by H_0 recovers its square-root energy relation. U_i is an ordered motion object, not a renamed Hermitian velocity. Replacing its positive square by ordinary Heisenberg velocity squared fails at finite Landau spacing, with residuals of either sign. The ordered identity has the correct fixed-kinetic-ratio classical limit. Its adjacent-level frequency is

$$\omega = \frac{2Q_0|e|B}{m(Q_{m,n} + Q_{m,n+1})}. \tag{25}$$

This agrees with the independently rationalized Landau energy difference and uses a harmonic-mean clock factor. The positive expectation identity separates ordered mean motion, mean clock progression and variances. The admitted commutators and charged quantum representation remain inputs; spin, elementary charge and transition rates have not been derived.

The second extension allows a spatially nonuniform, noncommuting clock operator. Hold A and B fixed in the rest-energy derivative and let $H(E_0)=\sqrt{(E_0^2A+B)}$, with $0\le A\le I$ and $B\ge 0$. Ordinary scalar differentiation is invalid. Instead the derivative clock solves a Lyapunov equation:

$$\begin{gathered} H\beta + \beta H = 2E_0 A, \qquad 0 \le \beta \le \|\sqrt{A}\| I \le I, \\ H = E_0\beta + M, \qquad M \ge 0. \end{gathered} \tag{26}$$

A positive inverse for that equation gives the clock bound; the corresponding finite-increment bound does not require a spectral gap. The apparently stronger local ordering $\beta \le \sqrt{A}$ is false, with an explicit two-dimensional counterexample. The derivative clock of a delocalized state cannot be identified with multiplication by a local lapse. If the source itself changes A or B when E_0 changes, their derivatives must be included; an added electrostatic potential likewise needs a fresh positivity check.

$$0 \le H(E_2) - H(E_1) \le \|\sqrt{A}\|(E_2 - E_1)I \qquad (E_2 \ge E_1). \tag{27}$$

An actual packet provides more than a matrix identity. For a supplied static one-dimensional lapse, the weighted Klein-Gordon operator has measure dx/N. Transforming to ordinary dx retains every derivative term and yields

$$H^2 = E_0^2 N^2 + C^\dagger C, \qquad C = \hbar c \sqrt{N}\, \partial_x \sqrt{N}. \tag{28}$$

With $N = 0.55 + 0.35\cos x$, $E_0 = c = 1$ and $\hbar = 0.12$, unitary packet evolution conserved H at 0.83442606199856 while the clock share $E_0\langle\beta\rangle$ ranged approximately from 0.07218 to 0.75618. The positive complementary share changed oppositely. The 192-to-256 point history difference was 4.88×10^{-12}; coarse 64/96-point histories failed the long-time refinement check. An independent weighted-operator calculation agreed at sampled times within 2.4×10^{-14}. This is energy exchange in an admitted external quantum background, without source backreaction.

Finally, the retained classical Hamiltonian makes the connection back to capacity explicit. For $H = N\sqrt{E_0^2 + c^2 q^{ij} p_i p_j}$, the energy derivatives give $\beta = N^2 E_0/H$ and $V^i = N^2 c^2 q^{ij} p_j/H$. Consequently

$$\begin{gathered} H = E_0\beta + p_i V^i, \qquad S^2 + q_{ij}V^iV^j + c^2\beta^2 = c^2, \\ H\beta = E_0 N^2 = E_0\beta^2 + mV_{\mathrm{rel}}^2. \end{gathered} \tag{29}$$

The last quadratic partition is a different expression from the canonical energy itself. A massive probe's Q_0/β is also distinct from the background source's held reading Q_s/N: the former is singular in a fixed-photon-energy zero-mass limit. Photons are propagated by H_ɣ=Nc|p|_q without assigning them a rest-source clock. These distinctions leave a coherent next step: use the same complete energy to evolve actual signed matter, then calculate the field organization and its observable magnetic share.

10. Signed material motion is the source of a magnetic moment

A rotating neutral source can possess a magnetic field. The important material quantity is the signed relative motion of its charged constituents. In the retained weak-motion limit, the current is the charge-weighted sum of their velocities. For a localized stationary current, its dipole moment is the first rotational moment of that current:

$$\begin{aligned} \mathbf{j} &= \sum_s q_s n_s \mathbf{v}_s, \qquad \nabla\cdot\mathbf{j} = 0, \\ \mathfrak{m} &= \tfrac{1}{2}\int \mathbf{r}\times\mathbf{j}\, dV = \sum_s \tfrac{q_s}{2m_s}\mathbf{L}_s. \end{aligned} \tag{30}$$

Here q_s and m_s are the charge and inertial mass of a material species, n_s is its number density, and L_s is its mechanical angular momentum about the stated origin. This orbital-current identity assumes a fixed charge-to-mass ratio within each species. An intrinsic spin magnetization contributes its own retained current, curl M. Neither electric charge nor that spin response is identified with SET's positive source rate Q_s.

For two species with charges +e and -e and identical number-density profiles, common rigid rotation cancels from j exactly. Relative rotation survives. This explains why knowing total mass, radius and bulk rotation does not yet specify the current that an electromagnetic field reads. It also explains why a neutral object can be magnetic without carrying a net electric charge.

The same source work reaches the two species differently

The common-energy charged equations reduce locally to the material momentum balance below. P_s is the species pressure tensor and R_s its collisional momentum exchange. These material quantities require an equation of state and transport physics.

$$m_s n_s D_s \mathbf{v}_s = m_s n_s \mathbf{g} + q_s n_s(\mathbf{E} + \mathbf{v}_s\times\mathbf{B}) - \nabla\cdot \mathrm{P}_s + \mathbf{R}_s \tag{31}$$

Subtract the acceleration equations for singly charged positive and negative populations. The direct universal gravitational acceleration cancels from the relative acceleration, while electric response, magnetic turning, pressure differences and collisions remain. Gravity still shapes the density, pressure, phase separation and convection from which the relative motion develops. It is therefore useful to calculate the mechanical work of the source before assigning any magnetic amplitude.

In the stated resistive macroscopic reduction, the equations include

$$\begin{aligned} \partial_t \mathbf{B} &= \nabla\times(\mathbf{u}\times\mathbf{B}) - \nabla\times(\eta\nabla\times\mathbf{B}) + \mathbf{C}_{\mathrm{batt}}, \\ \eta &= (\mu_0\sigma)^{-1}, \qquad \nabla\cdot\mathbf{B} = 0. \end{aligned} \tag{32}$$

The velocity u is the material bulk flow, σ is conductivity, and C_batt denotes a separately specified material battery if one is retained. The exact zero-field state remains zero under the homogeneous deterministic induction equation with homogeneous magnetic boundary

conditions. A seed or battery must have its own source and energy accounting. Rotation can organize the flow; the Coriolis force itself does no work. [N8-N10.]

11. Burden gradients supply calculable material work

In the retained spherical weak-field branch, the source-energy selection gives the symmetric potential kernel and $S^2=-2\Phi$. With the potential zero at infinity, a spherical material profile supplies

$$M(r) = 4\pi\int_0^r \rho(s)s^2\,ds,$$
$$\Phi(r) = -G\left[\frac{M(r)}{r} + 4\pi\int_r^R \rho(s)s\,ds\right], \qquad S^2(r) = -2\Phi(r). \tag{33}$$

Exterior material shells contribute to the interior clock and burden even though their spherical force cancels there. A boundary source Q divided by every interior area would lose this distinction. The force and the gravitational self-energy are

$$\mathbf{g} = \tfrac{1}{2}\nabla S^2, \qquad W_g = \tfrac{1}{2}\int \rho\Phi\,dV = -\tfrac{1}{4}\int \rho S^2\,dV,$$
$$\delta W_g = \int \Phi\,\delta\rho\,dV = -\tfrac{1}{2}\int S^2\delta\rho\,dV. \tag{34}$$

The last equality varies both the material and its sourced field. The two source-kernel contributions are equal, removing the factor one half in the variation. This is the direct link from a change of material organization to available gravitational work. It does not require treating Q, which has units of volume per time, as an additional power supply in watts.

A worked terrestrial source

Integrating the published PREM density profile without adjusting its mass gives $M=5.97318\times10^{24}$ kg and a binding energy of -2.48892×10^{32} J. Independent kernel and radial-energy integrations agree numerically. The source boundary gives $Q=5.70613\times10^{18}$ m³/s. These are the weak spherical readings of the specified model. [M1; N9.]

Location	g (m/s²)	S (m/s)	Clock deficit (ns per reference second)
Centre	0	14,951.1	1.2436
Inner-core boundary	4.4031	14,768.7	1.2134
Core-mantle boundary	10.6893	13,537.2	1.0195
Surface	9.8219	11,187.1	0.6962

For the stated linear composition model, light material rejected during freezing is redistributed through the liquid outer core. Let βχ be its compositional expansion coefficient and Δχ the rejected light-element fraction. The mass-weighted receiving-region average gives the work per unit freezing mass:

$$\varepsilon_g = \frac{\beta_\chi \Delta\chi}{2}\left[S_{\mathrm{ICB}}^2 - \langle S^2 \rangle_{\mathrm{outer}}\right] \tag{35}$$

With the independently supplied geometry and oxygen-partition inputs, ε_g=0.27886 MJ/kg. The comparison material calculation gives 0.254 MJ/kg. The difference is retained. Once the phase-change rate is determined from the thermal and composition equations, multiplying by that rate supplies a mechanical-work scale; it does not yet supply a dipole fraction. [M1-M3; N9.]

$$P_b = \frac{1}{2}\int \delta\rho\, \mathbf{u}\cdot\nabla S^2\, dV \tag{36}$$

This is the buoyancy power after the hydrostatic reference has been removed. Upward-moving light material has positive P_b in the retained terrestrial field. The source law and capacity reading therefore connect to an explicit work integral before any magnetic observation is introduced.

12. Thermal, phase and magnetic energy must share one source ledger

The updated material calculation enforces one temperature at the core-mantle boundary. The retained PREM adiabat uses d ln T/dr=-γg/v_bulk², with γ=1.5 and an inner-core boundary temperature of 5700 K. In the solid, v_bulk²=V_P²-4V_S²/3. Its outer-core boundary temperature is 4185.97 K. [M1-M3; N10.]

$$\begin{aligned} T_{\mathrm{core}}(R_c) &= T_{\mathrm{mantle}}(R_c) = T_c, \\ -k_{\mathrm{core}} T'_{\mathrm{core}} &= -k_{\mathrm{mantle}} T'_{\mathrm{mantle}} = q_{\mathrm{CMB}}. \end{aligned} \tag{37}$$

The published mantle boundary-layer response, with its material normalization kept fixed, gives a heat extraction of 18.06094 TW for this temperature branch. The corresponding adiabatic conductive contribution is 16.81504 TW. The 1.24590 TW difference is a boundary thermal excess. The larger gravitational and latent contributions must still be obtained from the phase and energy balances. This mantle law is an external constitutive input, not a calibration to the magnetic field. [M3-M4.]

$$q_{\mathrm{CMB}} = 0.0894\left[\frac{T_c - 2692\,\mathrm{K}}{3900\,\mathrm{K} - 2692\,\mathrm{K}}\right]^{4/3} \mathrm{W\,m^{-2}}, \qquad H_{\mathrm{CMB}} = 4\pi R_c^2 q_{\mathrm{CMB}}. \tag{38}$$

For cooling x=-dT_c/dt, differentiate the melting condition and the source-energy integrals. In the local quasistatic material approximation,

$$\dot{r}_i = C_{ri}x, \qquad \dot{M}_i = 4\pi r_i^2 \rho_s C_{ri} x,$$
$$H_{\mathrm{CMB}} = (C_s + C_L + C_g)x, \quad C_L = L_f \frac{\dot{M}_i}{x}, \quad C_g = \varepsilon_g \frac{\dot{M}_i}{x}. \tag{39}$$

L_f is the latent heat per unit mass. The derivative C_ri comes from the liquid-side adiabat and liquidus slopes, including composition when retained. Two explicitly supplied phase-response choices give the following results at the same thermal interface; they are separate material branches.

Quantity	Frozen composition derivative	Dilute Fe-O derivative
Cooling (K/Gyr)	136.18	175.17
Inner-core growth (mm/year)	1.4147	1.1570
Compositional work (TW)	2.9917	2.4467
Available entropy (MW/K)	871.26	719.09
Outer-core Ohmic allowance (TW)	4.9662	4.0988

The Ohmic entries allocate all available entropy to the specified outer-core dissipation and therefore are allowances. Other dissipative channels reduce them. Their numerical integration converges much more tightly than the uncertainty in composition and transport. The phase choices, not numerical roundoff, set the meaningful spread. [N10.]

$$P_{\mathrm{Ohm,\,out}} \le T_i \dot{S}_{\mathrm{available}}, \qquad T_i = 5700\,\mathrm{K}. \tag{40}$$

Finite cooling keeps the energy of departures from the reference

$$\mathcal{H}_{\mathrm{complete}} = H_{\mathrm{ref}}(T_c) + E_{\mathrm{dev}} + K + U_B, \qquad H'_{\mathrm{ref}} = C_{\mathrm{eff}},$$
$$-\dot{T}_c = \frac{H_{\mathrm{out}} + \dot{E}_{\mathrm{dev}} + \dot{K} + \dot{U}_B - P_{\mathrm{ext}}}{C_{\mathrm{eff}}}. \tag{41}$$

E_dev contains the thermal layer, unmixed composition and corresponding gravitational departure from the reference family. This identity follows by differentiating the complete energy and retaining exported heat. It specifies how a finite material simulation must feed back into its mean cooling. Removing those derivatives while separately evolving their energies would break the common accounting.

A separate two-reservoir wall-diffusion control debits adiabatic flux from the cooling background and excess flux from the boundary layer. It is not the single-adiabat cooling trajectory in the table. With the same boundary response, it gives wall-temperature perturbations relative to that background of -0.00188 K after one year, -0.01884 K after one hundred years and -0.05956 K after one thousand years. The layer energy loss matches the excess export. This is the

low-frequency Fourier approximation with thermal feedback; convection, rotation and material instability are separate evolution questions. The full 18.06 TW mean-feedback prescription was not installed in a completed terrestrial dynamo calculation. [N10.]

13. What the material and induction calculations establish

The coupled calculation keeps kinetic work, magnetic storage, dissipation and heat return in the same energy. The useful local balances are

$$\begin{aligned} \dot{K} &= P_b - W_L - D_{\text{visc}} + P_{\text{boundary}}, \\ \dot{U}_B &= W_L - D_{\text{Ohm}} - F_{\text{out}}, \\ W_L = -\int \mathbf{u}\cdot(\mathbf{j}\times\mathbf{B})\,dV, &\qquad D_{\text{Ohm}} = \int \frac{j^2}{\sigma}\,dV. \end{aligned} \tag{42}$$

The Lorentz work leaves the material reservoir and enters the magnetic one. Ohmic and viscous losses reappear as heat when the material reservoir is included. F_out denotes the explicitly retained outward electromagnetic flux. This reciprocal structure is the energy-selection result applied to an evolving conductor.

The matched conductivity and temperature profiles give a dipole decay time of 58,182 years. A fluctuation-dissipation seed calculation for one real dipole orientation gives an effective temperature of 4933.5 K and a surface-equatorial RMS field of 1.126×10^{-24} T. Its mean signed field is zero. These results use declared thermal fluctuation physics and an independently solved mode; they supply a seed ensemble and decay scale, not an assumed geomagnetic amplitude. [N10.]

The numerical controls establish that the retained nonlinear exchange can be implemented. A six-wavevector Galerkin test closes its ideal nonlinear energy exchange to roundoff and returns viscous and resistive loss to heat. A spherical dynamo benchmark reproduces its published kinetic and magnetic energies to 0.00287% and 0.00108% at the matched grid. A variable-conductivity test reproduces 51 reference output rows at their printed precision. [N10; M5.]

Changing to the independently specified conductivity profile at a common short comparison time changes outer magnetic energy by -0.1431%, kinetic energy by +0.0776% and complete-core Joule power by +4.7074%. These are computed responses of the coupled equations. The numerical regime still uses $E=10^{-4}$ and Pr=Pm=1, with stated perturbations. Its magnetic diffusion-to-rotation ratio is about 59,175 times the terrestrial reference. A dimensional rescaling of its field would not make it an Earth-strength prediction.

An unequal-conductivity audit also located a missing reciprocal-conductivity factor in an upstream Joule diagnostic. Correcting that diagnostic left the evolved checkpoints unchanged and restored the intended energy comparison. The final short-run balance residual is 0.01097% of integrated absolute driving and loss. An earlier underresolved later-time run is excluded from the accepted controls. [N10.]

The complete source can be constant while magnetism evolves

$$2\frac{dQ_s}{Q_s} - \frac{dV_{\mathrm{ref}}}{V_{\mathrm{ref}}} = \frac{d\mathcal{H}_{\mathrm{complete}}}{\mathcal{H}_{\mathrm{complete}}} \tag{43}$$

At fixed reference volume, an internal transfer with dH_complete=0 leaves Q_s unchanged. The magnetic field can grow, decay, change geometry or return energy to heat during that transfer. A conserved complete source therefore constrains all reservoirs together. It does not freeze one reservoir or its dipole component. The next derivations calculate how the field geometry determines that component.

14. The rotational clock expression and its magnetic reading

The planetary calculation began with a simple capacity difference. For the stated spherical boundary, take the total source mass M, mean radius R and sidereal angular speed Ω. The held and moving clock readings differ because the boundary motion consumes capacity. Retaining the same speed convention throughout,

$$\begin{gathered} Q = 4\pi\sqrt{2GMR^3}, \quad \bar{\rho} = \frac{3M}{4\pi R^3}, \quad S = \frac{Q}{4\pi R^2}, \quad v = \Omega R, \\ \alpha^2 = 1 - \frac{S^2}{c^2}, \qquad \beta^2 = \alpha^2 - \frac{v^2}{c^2}, \qquad \alpha - \beta = \frac{v^2}{c^2(\alpha+\beta)}. \end{gathered} \tag{44}$$

Multiplication by c√(μ₀ρ̄) expresses this clock difference in tesla. Call the resulting algebraic amplitude b₀, to distinguish it from an independently calculated magnetic field. The positive energy remainder associated with that amplitude is

$$\begin{gathered} b_0 = c\sqrt{\mu_0\bar{\rho}}\,(\alpha-\beta) = \frac{\sqrt{\mu_0\bar{\rho}}\,v^2}{c(\alpha+\beta)}, \\ U_* = Mc^2(\alpha-\beta)^2. \end{gathered} \tag{45}$$

The rationalized expression avoids subtracting almost equal clock factors numerically. The units and energy identity are exact. A pure exterior dipole with equatorial amplitude b at R has U_d,out=4πR³b²/(3μ₀). Therefore setting b=b₀ is exactly the extra assignment U_d,out=U*. This is the assignment tested against the observations below. It is not needed to derive the scalar identity. [N11-N14.]

The same remainder in clock and motion energies

$$\begin{gathered} \Delta C = Mc^2(\alpha-\beta), \quad K_0 = \frac{Mv^2}{2}, \quad K_{\mathrm{can}} = \frac{Mc^2\alpha(\alpha-\beta)}{\beta}, \\ U_* = 2(\alpha\Delta C - K_0) = \beta(K_{\mathrm{can}} - \Delta C). \end{gathered} \tag{46}$$

These equalities identify one positive remainder in several invariant-derived readings. They do not say that it is the entire energy available to every current. In the weak fields of the listed bodies,

$$b_0 \simeq \frac{\Omega^2}{2c}\sqrt{\frac{3\mu_0 MR}{4\pi}}. \tag{47}$$

G cancels at this order. The capacity expression remains useful as a compact trial scale, but a large empirical correction cannot be assigned to ordinary gravitational clock normalization in these weak fields. In particular, with u=S²/c² and fixed mean density and v, ∂ ln b₀/∂u=1/(2αβ)>0. Increasing burden raises this candidate rather than reducing its large overestimates. Rewriting the same expression using Q/α changes no prediction when every factor is transformed consistently.

15. What the unchanged planetary and solar calculation gives

Earth’s measured field was known when this candidate was identified. Its bulk inputs and formula were then frozen for the transfer calculations; Jupiter was the predeclared unchanged transfer test. The measured planetary quantity is the magnitude of the degree-one internal field at that same radius, not a maximum surface field or a field at the poles. Published Gauss coefficients with reference radius a are converted by

$$b_1(R) = \sqrt{(g_1^0)^2 + (g_1^1)^2 + (h_1^1)^2}\,\left(\frac{a}{R}\right)^3. \tag{48}$$

Body	Candidate b₀ (μT)	Dipole comparison (μT)	b₀ / comparison
Earth	29.963126	29.737833	1.00758
Mercury	0.00126046	0.190140	0.006629
Jupiter	10,290.638	446.640438	23.0401
Saturn	4,457.926	23.436044	190.2167

Earth is a close result: 0.758% above the retained WMM2025 dipole. It is preserved as calculated. Jupiter and Saturn are overestimated, while Mercury is underestimated. The assignment does not transfer as a universal bulk mass-radius-rotation law. Including all retained WMM harmonics gives an exterior-energy-equivalent amplitude 30.333939 μT, for which b₀ is 1.22% low; this is a different observable from the dipole. The coefficients, comparison radii and raw inputs are preserved in the research records. [D1-D7; N11-N14.]

The solar comparison uses a different explicitly stated observable: the photospheric axial harmonic converted to the vacuum dipole-equatorial equivalent |D_ax|/2. It is not the full tilted dipole or a sunspot field. The unchanged candidate is 2.789484 gauss. The five published WSO minima give the following comparison. [D8.]

Solar minimum	Axial-equivalent observation (G)	Candidate excess
1976	1.975	41.24%
1986	2.520	10.69%
1996	1.895	47.20%
2008	0.875	218.80%
2019	1.095	154.75%

The nearest solar case is worth keeping, together with the full series. A separate angular-current trial uses the published differential rotation $\Omega(x)=14.37-2.33x^2-1.56x^4$ degrees/day, x=sin(latitude), and the declared local weighting jφ proportional to r sinθ $\Omega(x)^2$. Its angular integral gives a factor 0.945075 and a candidate 2.636271 G, 4.61% above the 1986 WSO value. Extending that rotation law through the chosen volume and selecting this current weighting are extra trial prescriptions; the source law has not selected them. No epoch was used to fit the factor. [N13.]

Mars and Venus supply useful independent diagnostic limits. The same bulk expression gives 6768.85 nT for Mars against a published intrinsic global-dipole limit below 0.5 nT. Crustal fields are a different observable. For Venus it gives 0.44397 nT against a named historical limit of 0.37899 nT. The latter is an upper-limit comparison, not a detected agreement or a quantified significance. Both objects were chosen as weak-field controls before evaluating the candidate. [D9-D10; N14.]

16. Measurement history and physical source history

A specific possibility was tested: perhaps the mismatch increases because distant bodies have been measured less thoroughly. The appropriate comparison is what improved magnetometry actually changed, after matching reference radii. Twenty giant-planet model records and sixteen Mercury entries were audited. Retrospective fits and shared mission data were not treated as independent samples. [N14.]

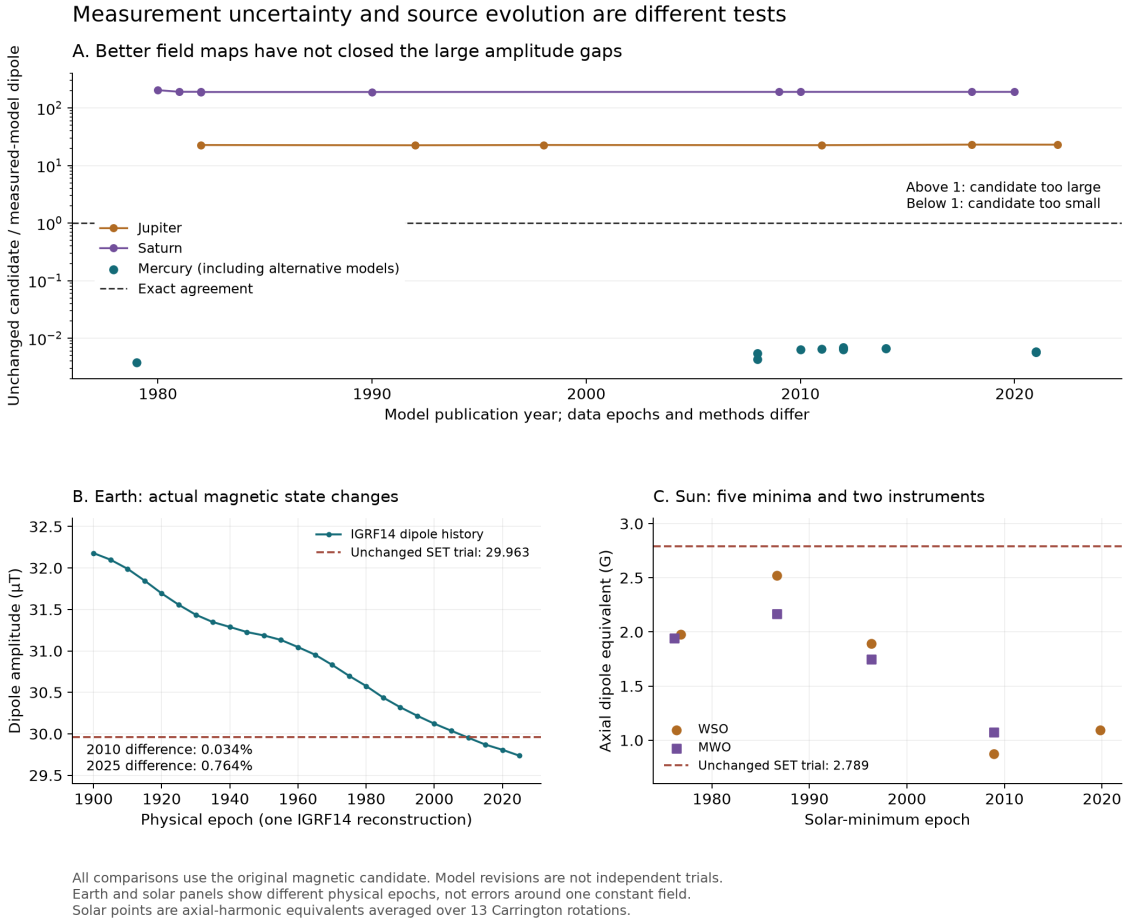
Body	Earlier common-radius dipole	Later common-radius dipole	Candidate
Jupiter	453.150 μT (Voyager)	446.640 μT (JRM33)	10,290.638 μT
Saturn	21.8775 μT (Pioneer)	23.4369 μT (Cassini11+)	4,457.926 μT
Mercury	188.139 nT (common-method old-data fit)	190.140 nT (orbital fit)	1.2605 nT

Saturn moves toward the candidate in this early-to-late comparison, but its 7.13% measurement revision closes only 0.0352% of the original absolute gap. Jupiter remains near its earlier dipole scale. Mercury has genuine inversion uncertainty: later full-mission alternatives give approximately 215-223 nT at the comparison radius. Those alternatives increase the discrepancy

with 1.26 nT. The available measurement history does not supply the changes needed to repair the universal assignment. This does not require an arbitrary ranking of human research effort. [D4-D7.]

Earth provides a different test. IGRF14 reconstructs changing physical field states: at the fixed comparison radius its dipole falls from 32.1792 μT in 1900 to 29.7360 μT in 2025. The frozen-input candidate stays at 29.9631 μT and passes very close to the 2010 value, 29.9528 μT. That closer epoch does not replace the original WMM2025 comparison. A near match at one time and the observed time history are distinct questions. [D3.]

The solar series makes the same distinction. The WSO minima vary by a factor 2.88. Paired MWO readings differ by roughly 8-19% at the listed epochs, so calibration matters to the nearest agreement. The variation across minima cannot be used as an uncertainty interval for one constant source state. The published table does not supply the covariance needed for a formal probability of agreement. [D8.]



Measurement and state controls for the unchanged rotational-clock candidate. Historical models are compared at a common radius; Earth's IGRF curve describes physical secular evolution, while the solar points describe different minima. A wider historical or temporal range is not automatically an observational error bar. Data and observable conventions are retained in [N14] and [D3-D8].

A new energy reading of the terrestrial history

$$U_{d,\,\mathrm{out}}(R) = \frac{4\pi R^3 b_1^2}{3\mu_0},$$
$$\Delta U_{d,\,\mathrm{out}} = \frac{4\pi R^3}{3\mu_0}\left[b_1(t_2)^2 - b_1(t_1)^2\right]. \tag{49}$$

IGRF epoch	External dipole energy (J)
1900	8.92592×10^{17}
2010	7.73355×10^{17}
2025	7.62201×10^{17}

Using the same radius 6,371,008.4 m gives a 1900-2025 change of -1.30390×10¹⁷ J, an average -33.05 MW in this component. Over 2010-2025 it is -23.56 MW. This is the vacuum-equivalent energy of the degree-one internal field outside the stated sphere. It is not the total dynamo power, Ohmic heat or radiated power. Its importance for SET is direct: an internal exchange can change this observable while leaving the complete source energy, and hence Q_s at fixed reference volume, unchanged.

17. Testing source organization without a fitted efficiency

The current geometry matters before the magnetic energy is observed at an outer boundary. As a trial, apply the same rotational-clock assignment to an independently specified active mass M_a inside radius a and continue its dipole to R. Keeping its clock normalization explicit gives

$$\frac{b_R}{b_0} = \sqrt{\frac{M_a}{M}}\left(\frac{a}{R}\right)^{7/2}\left(\frac{\Omega_a}{\Omega_R}\right)^2 \frac{\alpha_R + \beta_R}{\alpha_a + \beta_a}. \tag{50}$$

The exponent 7/2 follows from the trial’s source and dipole algebra. The choice of active domain and the assignment of clock remainder to its external dipole remain supplied. An embedded domain’s clock includes the surrounding gravitating material. The weak-field numerical tests do not repair that input by treating the inner domain as an isolated gravitating object.

Independent material trial	Surface candidate	Comparison
Earth: PREM liquid outer core	2.0038 μT	29.7378 μT
Mercury: independently constrained core	0.5565 nT	about 190 nT
Saturn: metallic shell, stated mean-radius mapping	987.175 μT	23.436 μT

The Saturn domain spans 0.37-0.733 model radii and has mass fraction 0.4313 calculated by integrating its tabulated density under the stated spherical mapping. Its independent material model is not a direct measurement of a sharp dynamo boundary. The change reduces the original factor 190 discrepancy to about 42. Earth and Mercury move farther from their

observations. Keeping source structure is physically necessary, but these trials do not select one universal correction. A separately declared family of twenty algebraic alternatives also left no common expression within a factor of two for all four original planetary comparisons. [M1; D11; N13-N14.] The independent Mercury core inputs use Hauck et al. (2013) and Margot et al. (2012).

A useful algebraic diagnostic is the energy ratio that the original identification would require. U_observed/U* is 0.9850 for Earth, 0.0018838 for Jupiter, 0.00002764 for Saturn and about 22,756 for Mercury. These numbers are not inserted into a new formula. Their mixed range shows that a passive fraction between zero and one cannot by itself repair every object if U* is retained as the whole available budget. The complete material energy and signed current preparation must be calculated.

Total magnetic energy differs from the observed exterior

For a dipole generated inside Earth's core radius R_c, the spherical field bound gives $U_B \geq 3(R/R_c)^3 U_{d,out}(R)$, about 18.41 times the surface-exterior energy for the retained core radius. A regular declared current profile $j = k\, e_z \times r$ in the outer-core shell, normalized to the candidate dipole, instead gives complete $U_B = 2.02199 \times 10^{19}$ J, 26.13 times U*. Its calculated Ohmic loss is 24.25 MW. This is a field completion of the supplied candidate amplitude, not a solution that generated that amplitude from core evolution. It makes the missing energy accounting explicit. [N11-N13.]

The stronger result of the present work is to calculate these allocations from specified current dynamics and spatial variation. The next sections derive their geometry, a sharp constrained energy bound and a finite-driver conversion experiment. The original Earth and solar values remain in the record without being used to choose those calculations.

18. Projecting the common energy onto magnetic modes

The retained source law makes one complete energy the generator of clocks, material motion and field response. To calculate energy in an organized magnetic mode, that generator must be evaluated on a specified charged state. A numerical source reading is one value of the functional; it does not replace its signed current variables, conserved quantities or initial and boundary data.

$$\mathcal{Q} = \frac{Q_s^2}{V_{\mathrm{ref}}} = \frac{24\pi G}{c^2}\mathcal{H}, \qquad \dot{z} = \{z, \mathcal{H}\} = \frac{c^2}{24\pi G}\{z, \mathcal{Q}\} \tag{51}$$

For a fixed geometry with zero shift, expand a vector potential in independent physical modes A_n with amplitudes a_n. Write B_{dens}=curl A for the magnetic vector density, N for the lapse, q_{ij} for the spatial ruler and q for its determinant. The retained Maxwell energy defines a positive matrix after gauge zero modes have been removed:

$$\begin{aligned} \mathcal{H}_B &= \tfrac{1}{2}\mathbf{a}^T \mathsf{K} \mathbf{a}, \\ K_{nm} &= \tfrac{1}{\mu_0}\int \frac{N q_{ij}}{\sqrt{q}} (\nabla \times A_n)^i (\nabla \times A_m)^j \, d^3x. \end{aligned} \tag{52}$$

Orthonormal modes in this energy metric give an additive positive partition. Nonorthogonal pieces retain their cross terms. For a self-gravitating source, the same variation must also retain the response of the ruler, material and support. A flat exterior field energy is consequently not substituted for the whole strongly bound source energy.

Specified canonical data can select an allocation

A finite slow-motion mode reduction makes the calculation explicit. Let P be cyclic carrier momenta, I their positive material inertia matrix, C their charge coupling to a, and p_a the electric conjugate with positive matrix M. All coefficients come from the supplied matter profiles and the same Maxwell field:

$$\mathcal{H} = H_0 + \tfrac{1}{2}(\mathbf{P} - \mathsf{C}\mathbf{a})^T \mathsf{I}^{-1}(\mathbf{P} - \mathsf{C}\mathbf{a}) + \tfrac{1}{2}\mathbf{a}^T \mathsf{K}\mathbf{a} + \tfrac{1}{2}\mathbf{p}_a^T \mathsf{M}^{-1}\mathbf{p}_a. \tag{53}$$

At specified conserved P, the stationary field has $p_a=0$ and a unique amplitude in this positive quadratic sector:

$$\begin{gathered}(\mathsf{K} + \mathsf{C}^T \mathsf{I}^{-1}\mathsf{C})\mathbf{a}_* = \mathsf{C}^T \mathsf{I}^{-1}\mathbf{P}, \\ \mathcal{H}_{B,*} = \tfrac{1}{2}\mathbf{a}_*^T \mathsf{K}\mathbf{a}_*.\end{gathered} \tag{54}$$

This is a conditional allocation derived from the common energy. Different admitted carrier momenta give different answers. The closed Hamiltonian also permits oscillations about this minimum; reaching it requires a stated physical process. A fixed total bulk angular momentum generally does not determine the relative motion of oppositely charged species. The following calculation solves a spatial organization problem without inserting an empirical efficiency.

19. The energy cost of an exterior magnetic harmonic

Assume a finite-energy divergence-free magnetic field in Euclidean space, constant μ_0, all assigned currents enclosed by r=R, and a current-free decaying exterior. Expand its normal boundary field as $B_r(R)=\Sigma q_{lm} Y_{lm}$, with $\int|Y_{lm}|^2 d\Omega=1$ and $l\geq 1$. The regular interior and decaying exterior potential fields for one harmonic are

$$\begin{gathered}\Phi_{\text{in}} = -\frac{q_{lm}}{l} R^{1-l} r^l Y_{lm}, \qquad \Phi_{\text{out}} = \frac{q_{lm}}{l+1} R^{l+2} r^{-l-1} Y_{lm}, \\ \mathbf{B} = -\nabla\Phi.\end{gathered} \tag{55}$$

For any other solenoidal interior with the same normal field, write $B=B_p+\delta B$. Integration by parts gives $\int B_p\cdot\delta B\, dV=0$ because div $\delta B=0$ and $\delta B\cdot n=0$. Its energy therefore exceeds the potential interior by $\int|\delta B|^2/(2\mu_0)$. Green's identity and harmonic orthogonality give

$$\begin{gathered}U_{\text{in},p,lm} = \frac{R^3|q_{lm}|^2}{2\mu_0 l}, \qquad U_{\text{out},lm} = \frac{R^3|q_{lm}|^2}{2\mu_0(l+1)}, \\ U_B \geq \frac{2l+1}{l} U_{\text{out},l}, \qquad \frac{U_{\text{out},l}}{U_B} \leq \frac{l}{2l+1}.\end{gathered} \tag{56}$$

Other harmonics and nonpotential interior fields only raise the total. Equality permits a surface current sheet supporting the tangential field jump. If only smooth volume currents are allowed, the same lower bound is approached by a thin current layer. For a dipole m, whose equatorial amplitude at R is b, this reproduces the previously recorded complete-field bound:

$$\begin{gathered} U_{\mathrm{out},\,d} = \frac{\mu_0 m^2}{12\pi R^3} = \frac{4\pi R^3 b^2}{3\mu_0}, \\ U_B \geq 3U_{\mathrm{out},\,d}, \qquad f_d(R) = \frac{U_{\mathrm{out},\,d}}{U_B} \leq \tfrac{1}{3}. \end{gathered} \tag{57}$$

The minimizing dipole interior is uniform. These conditions concern the isolated field assigned to the source; external currents, radiation, material permeability or strong spatial curvature require their own energy problem. The enclosing current radius is specified independently. The bound constrains complete magnetic storage and does not identify it with complete gravitating energy.

20. An exact constrained organization of a neutral current

Consider equal uniform number densities n of species +e and −e in a fixed ball of radius R. Their masses give $m_{red}=m_+m_-/(m_++m_-)$. Hold the density, radius and common centre-of-mass motion fixed by ideal constraints that do no work on the admitted current variations. A state-dependent support energy would change this problem. In the small-relative-speed, magnetostatic branch, $j=ne(v_+-v_-)$. Define $\kappa=m_{red}/(ne^2)$. The relative carrier energy and the entire magnetic field energy are

$$\begin{gathered} E_{\mathrm{rel}} = \tfrac{\kappa}{2}\int_{r<R} j^2\,dV, \qquad U_B = \tfrac{1}{2\mu_0}\int_{\mathbb{R}^3} B^2\,dV = \tfrac{1}{2}\int_{r<R} \mathbf{j}\cdot\mathbf{A}\,dV, \\ E[\mathbf{j}] = E_{\mathrm{rel}} + U_B, \qquad \nabla\times\mathbf{B} = \mu_0\mathbf{j}. \end{gathered} \tag{58}$$

Admit divergence-free currents tangent to the sphere and prescribe the dipole vector $m=m\,e_z$. Its value is a constraint for this calculation, not a generally conserved quantity inferred from Q. Varying $E-\lambda m$ over solenoidal currents gives $\kappa j+A-(\lambda/2)e_z\times r=\nabla\chi$. The minimizing toroidal sector has constant χ, giving $\kappa j+A=(\lambda/2)e_z\times r$. Spherical symmetry permits the minimizing toroidal form $A_\varphi=a(r)\sin\theta$, $j_\varphi=J(r)\sin\theta$. The radial equations are

$$\begin{gathered} \kappa J + a = \tfrac{\lambda r}{2}, \qquad k^2 = \tfrac{\mu_0}{\kappa}, \qquad x = kR, \\ a'' + \tfrac{2}{r}a' - \left(\tfrac{2}{r^2} + k^2\right)a = -\tfrac{k^2\lambda r}{2}. \end{gathered} \tag{59}$$

Regularity at the centre and matching to $a_{out}=\mu_0 m/(4\pi r^2)$ without a surface-current sheet impose $a'(R)+2a(R)/R=0$. They give

$$\begin{gathered} a(r) = \tfrac{\lambda}{2}\left[r - \tfrac{3i_1(kr)}{ki_0(x)}\right], \qquad J(r) = \tfrac{3\lambda i_1(kr)}{2\kappa k i_0(x)}, \\ i_0(z) = \tfrac{\sinh z}{z}, \quad i_1(z) = \tfrac{z\cosh z - \sinh z}{z^2}, \\ i_2(z) = \tfrac{(z^2+3)\sinh z - 3z\cosh z}{z^3}. \end{gathered} \tag{60}$$

The moment integral follows from $d[r^3 i_2(kr)/k]/dr = r^3 i_1(kr)$. Multiplying the stationary equation by j and integrating gives $2E = \lambda m$. Thus

$$\begin{aligned} m &= \frac{4\pi}{3}\int_0^R r^3 J(r)\,dr = \frac{2\pi\lambda R^3}{\mu_0}\frac{i_2(x)}{i_0(x)}, \\ E_{\min}(m) &= \frac{\mu_0 m^2}{4\pi R^3}F(x), \qquad F(x) = \frac{1}{g(x)}, \\ g(x) &= \frac{i_2(x)}{i_0(x)} = 1 + \frac{3}{x^2} - \frac{3\coth x}{x}, \qquad 0 < g(x) < 1. \end{aligned} \tag{61}$$

This is the unique global minimum in the declared positive quadratic current class. For any perturbation with the same dipole, the cross energy is $\lambda\delta m = 0$ and the excess is $\kappa\int|\delta j|^2/2 + \int|\delta B|^2/(2\mu_0) > 0$ unless $\delta j = 0$. Higher harmonics or poloidal additions cannot lower it. This establishes a constrained minimum, without assuming that dynamics relaxes to it.

A work bound and the full allocation

If a separately justified positive storage budget satisfies $E_{rel} + U_B \le W$, the minimum supplies a sharp bound. Equality can be attained in the quadratic class by scaling the minimizing profile, provided its carrier speeds remain within the declared small-speed regime:

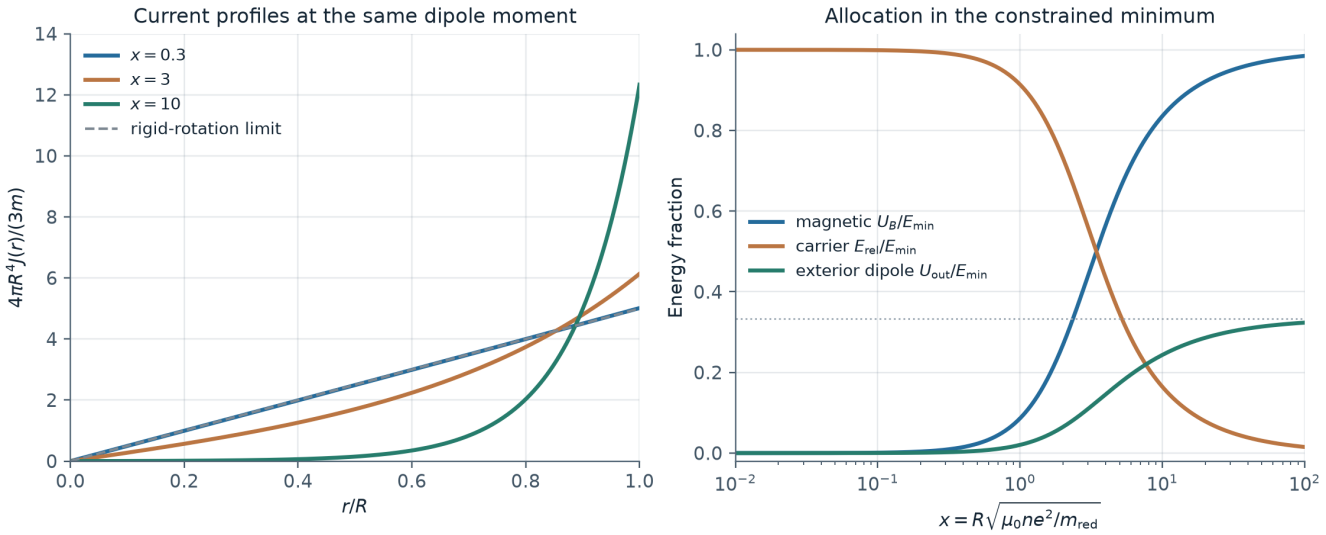
$$\begin{aligned} m^2 &\le \frac{4\pi R^3}{\mu_0} W g(x), \qquad b^2 \le \frac{\mu_0 W}{4\pi R^3} g(x), \\ U_{\mathrm{out},d} &\le \frac{g(x)}{3} W. \end{aligned} \tag{62}$$

Saturation has not been presumed. Other current patterns and other energy reservoirs can leave a smaller dipole. The relation $W = c^2\Delta(Q_s^2/V_{ref})/(24\pi G)$ may be inserted only if that comparison has already established that its complete energy increment is this positive storage budget. Changes of binding, support or internal energy must not be omitted to make that identification.

At the constrained minimum, the envelope derivative with respect to κ at fixed m, R, μ_0 separates carrier and field storage. With $C_m = \mu_0 m^2/(4\pi R^3)$,

$$\begin{aligned} E_{\mathrm{rel}} &= -\frac{C_m x F'}{2}, \qquad U_B = C_m\Big(F + \frac{xF'}{2}\Big), \qquad U_{\mathrm{out},d} = \frac{C_m}{3}, \\ \frac{E_{\mathrm{rel}}}{E_{\min}} &= \frac{xg'}{2g}, \quad \frac{U_B}{E_{\min}} = 1 - \frac{xg'}{2g}, \quad \frac{U_{\mathrm{out},d}}{E_{\min}} = \frac{g}{3}. \end{aligned} \tag{63}$$

As $x \to 0$, $F = 15/x^2 + 10/7 - x^2/147 + \ldots$, and the current approaches rigid relative rotation. Its magnetic energy tends to $5\mu_0 m^2/(14\pi R^3)$, with exterior fraction 7/30. As $x \to \infty$, current concentrates near the boundary, its carrier fraction vanishes and U_B approaches $3U_{out}$. These limits connect the regular uniform-rotation example to the sharp surface-layer geometry bound.



The exactly minimized neutral current at a prescribed dipole. Left: currents at the same m become concentrated near the boundary as x increases. Right: calculated carrier, total magnetic and exterior dipole shares; the exterior share is part of the magnetic share. No magnetic observation, fitted efficiency or dynamical relaxation assumption enters these curves.

Thirty-three exact checks and six independent current and field quadratures verify the radial equation, matching, moment, energy and limits; the largest numerical residual is 8.95×10^{-11}. This checks the stated approximation and derivation, not a universal material model or a planetary amplitude.

21. The complete source retains physical initial-state freedom

An explicit initial-data family shows what remains to be supplied. Start with a finite-energy admissible Maxwell state having pointwise zero net charge, E=0 and a divergence-free B_0. For a concrete admissible material realization, take two oppositely charged dust species, initially at rest with canceling local charge densities. Keep these mechanical data fixed. In the same physical frame set

$$\mathbf{E}_\theta = c\mathbf{B}_0 \sin\theta, \qquad \mathbf{B}_\theta = \mathbf{B}_0 \cos\theta. \tag{64}$$

Both Gauss constraints remain satisfied. The electromagnetic momentum vanishes, while the energy and every spatial stress component are identical:

$$\begin{aligned} \mathbf{E}_\theta \times \mathbf{B}_\theta = 0, \qquad E_\theta^2/c^2 + B_\theta^2 = B_0^2, \\ E_{\theta i}E_{\theta j}/c^2 + B_{\theta i}B_{\theta j} = B_{0i}B_{0j}. \end{aligned} \tag{65}$$

The same gravitational initial geometry and material stress therefore satisfy the same common-field constraints, with the same complete source energy. The magnetic storage nevertheless varies as $\cos^2\theta$. Regular examples can be constructed directly: choose a smooth divergence-free seed, a conformally flat spatial metric $q_{ij}=\psi^4\delta_{ij}$ and zero extrinsic curvature. For sufficiently small data, the positive nonlinear Hamiltonian-constraint integral equation is a contraction. It has the same solution ψ for every θ because its source is the unchanged

electric-plus-magnetic energy. The full derivation retains the solved boundary energy and its distinct gravitational weight.

This is ordinary physical initial-state freedom in the retained equations. With charged neutral matter it is an initial-data family, not a claimed symmetry of the whole subsequent solution: the electric force can produce different later motion. Holding mechanical velocities fixed requires the corresponding canonical momenta to shift with A. Stationarity, fixed magnetic flux, a specified current preparation and radiative boundary conditions can distinguish members; the source-energy equality alone does not supply those data.

The construction does not invent a nonspherical finite-screen Q law. It uses the retained complete-energy quotient, and any valid geometrical source reading is unchanged with the identical geometry. The constructive task is to supply or derive the charged preparation and its dynamics, calculate the organized field from the common energy, and then express that complete source in Q variables. The following worked preparations carry out that calculation in explicitly restricted current modes.

22. A regular current mode with a calculated magnetic share

A concrete example shows how source energy can be allocated once the signed material state is supplied. Take equal uniform number densities n of positive and negative carriers inside a supported sphere of radius a. Their charges are +e and -e, and their masses are m_+ and m_-. Each population is restricted to a rigid angular mode about the same axis. The profiles, mode restriction and preparation are explicit test data; the electromagnetic coupling and field energy are the retained ones.

Let N_p=4πna³/3 be the particle count of each species and q_e=eN_p the magnitude of each population's charge. The total local charge vanishes. The relative angular velocity $\omega=\Omega_+-\Omega_-$ nevertheless gives

$$j_\varphi(r,\theta)=\rho_e\omega r\sin\theta, \qquad \rho_e=en,$$
$$\mathfrak{m}=\frac{4\pi}{3}\int_0^a \rho_e\omega r^4\,dr=\frac{q_e a^2\omega}{5}. \tag{66}$$

The spherical current is divergence-free and tangent to the boundary. Write Aφ=f(r)sinθ. Ampere's equation becomes a radial equation with a regular origin and the decaying exterior dipole:

$$f''+\frac{2}{r}f'-\frac{2f}{r^2}=-\mu_0\rho_e\omega r,$$
$$f(r)=\mu_0\rho_e\omega\left(\frac{a^2 r}{6}-\frac{r^3}{10}\right), \quad 0\le r\le a, \tag{67}$$
$$f(r)=\frac{\mu_0\mathfrak{m}}{4\pi r^2}, \quad r\ge a.$$

Both the normal and tangential magnetic fields match at a. There is no magnetic surface-current sheet in this example. Integrating the field energy, or independently evaluating one half of the current-potential integral, gives

$$U_B = \int \frac{B^2}{2\mu_0}\, dV = \frac{1}{2}\int \mathbf{j}\cdot\mathbf{A}\, dV = \frac{\mu_0 q_e^2 a\omega^2}{70\pi},$$
$$U_{d,\,\mathrm{out}}(a) = \frac{\mu_0 \mathfrak{m}^2}{12\pi a^3} = \frac{7}{30}U_B. \tag{68}$$

The factor 7/30 has been calculated from the actual current geometry. It was not selected from the observed field. This gives a regular-volume example below the sharp one-third exterior bound.

Material backreaction determines the rest of the partition

Separate the common rotation from the signed relative mode. The material inertias are $I_+ = 2N_pm_+a^2/5$ and $I_- = 2N_pm_-a^2/5$. At fixed total material angular momentum, the quadratic kinetic energy splits into common rotation and a relative contribution.

$$K = \frac{L^2}{2(I_+ + I_-)} + \frac{1}{2}I_r\omega^2,$$
$$I_r = \frac{I_+ I_-}{I_+ + I_-} = \frac{2}{5}N_p m_{\mathrm{red}} a^2, \qquad m_{\mathrm{red}} = \frac{m_+ m_-}{m_+ + m_-}. \tag{69}$$

The field contributes an inductive inertia I_B. In the retained magnetoquasistatic reduction, eliminating the field consistently from the Lagrangian gives the canonical relative momentum P=(I_r+I_B)ω and its common Hamiltonian:

$$I_B = \frac{\mu_0 q_e^2 a}{35\pi}, \qquad P = (I_r + I_B)\omega,$$
$$H_{\mathrm{mode}} = \frac{P^2}{2(I_r + I_B)} = \frac{1}{2}(I_r + I_B)\omega^2. \tag{70}$$

The positive Hessian is 1/(I_r+I_B). A finite impulse must supply the canonical momentum including the induced field response. Computing the material speed from I_r alone while also adding the magnetic energy would overcount the delivered energy. Within this specified mode, the shares are now fixed:

$$\frac{U_B}{H_{\mathrm{mode}}} = \frac{I_B}{I_r + I_B}, \qquad \frac{K_r}{H_{\mathrm{mode}}} = \frac{I_r}{I_r + I_B},$$
$$\frac{U_{d,\,\mathrm{out}}}{H_{\mathrm{mode}}} = \frac{7}{30}\frac{I_B}{I_r + I_B}. \tag{71}$$

This is an actual allocation formula. Its ingredients are the specified carrier masses, density, geometry and charged coupling. There is no free magnetic efficiency. A different material mode has a different calculable response; the complete source energy by itself does not prescribe this mode.

A numerical calculation without an observed magnetic target

Choose a=1 cm, n=10^{20} m^{-3} and a supplied mode excitation of 10^{-18} J. Use measured proton and electron masses and the standard vacuum calibration. This is an ideal supported two-fluid verification problem, not a planet or a prediction of an independently observed laboratory plasma. [D12.]

Calculated quantity	Result
Relative material inertia	1.52546×10^{-20} kg m²
Inductive inertia	5.14742×10^{-19} kg m²
Relative angular speed	1.94258 rad/s
Magnetic share of the excitation	0.971218
Exterior dipole share of the excitation	0.226617
Equatorial field at a	0.260740 nT
Largest surface speed divided by c	6.48×10^{-11}

The field, carrier kinetic energy and source work balance exactly in the reduced model. Ten symbolic checks verify the field equation, boundary matching, energy integral, kinetic split and common generator. The source-law conversion remains available for a specified complete reference volume: $\Delta(Q_s^2/V_{ref})=(24\pi G/c^2)\Delta H_{complete}$. During an internal transfer from a finite driver, its energy loss must be included; the complete source Q_s can remain constant while H_{mode} grows.

The magnetic field is quasistatic here. Its finite-frequency electric storage and outgoing radiation are higher-order corrections, not hidden reservoirs counted as zero in a supposedly exact relativistic solution. The next experiment retains an actual finite driver, reciprocal feedback and heat production, and checks the size of the omitted quasistatic terms.

23. Exact inductive geometry for a finite material class

A source-energy allowance becomes an actual magnetic allocation only after a material evolution is supplied. The following test provides one explicitly: two neutral pairs of oppositely charged spherical shell profiles, each coupled to a finite mechanical driver. The magnetic fields start at zero. No observed field, chosen magnetic efficiency, or prescribed current history enters. The geometry is exact within the fixed-profile magnetostatic class; the subsequent evolution is a controlled magnetoquasistatic reduction, not a full causal Maxwell solution.

For shell pair i, let each charge magnitude be q_i, its radius be a_i, and the relative angular velocity of its two species be ω_i. The net charge density cancels. Eliminating the common bulk motion at fixed bulk angular momentum leaves a positive relative inertia. The relative current and its dipole moment are

$$\begin{aligned} K_{phi,i} &= \frac{q_i \omega_i}{4\pi a_i} \sin\theta, \qquad m_i = \frac{q_i a_i^2 \omega_i}{3}, \\ I_i &= \frac{I_{+,i} I_{-,i}}{I_{+,i} + I_{-,i}}, \qquad K_{\mathrm{rel}} = \tfrac{1}{2} \boldsymbol{\omega}^T I \boldsymbol{\omega}. \end{aligned} \tag{72}$$

The field is uniform inside an individual shell and dipolar outside. Integrating the combined field over all space, including mutual energy and its sign, gives the symmetric inductive inertia matrix. For an enclosing radius R,

$$\begin{aligned} U_B &= \tfrac{1}{2} \boldsymbol{\omega}^T L \boldsymbol{\omega}, \qquad L_{ij} = \frac{\mu_0 q_i q_j \min(a_i, a_j)^2}{18\pi \max(a_i, a_j)}, \\ U_{\mathrm{out}}(R) &= \frac{\mu_0}{12\pi R^3} \left(\sum_i m_i \right)^2. \end{aligned} \tag{73}$$

This counts the cross energy once through the quadratic form. Adding separate isolated-shell energies would lose the interference. With ordered radii and cumulative signed moments M_j, a telescoping identity makes the organization penalty explicit:

$$\begin{aligned} M_j &= \sum_{i \le j} m_i, \qquad R = a_N, \\ U_B &= 3U_{\mathrm{out}}(R) + \frac{\mu_0}{4\pi} \sum_{j=1}^{N-1} M_j^2 \left(a_j^{-3} - a_{j+1}^{-3} \right). \end{aligned} \tag{74}$$

Hence the exterior fraction lies between zero and one third. The upper bound is attained by an outermost current shell alone. Opposite signed moments can cancel the exterior dipole while retaining positive interior magnetic energy. This is a geometric result, not a saturation efficiency. The continuous dipolar-current limit replaces the sum by 3 times the integral of $M(r)^2/r^4$; uniform-density relative rotation then gives an exterior fraction 7/30, consistent with the separate uniform-ball calculation.

24. A finite driver calculates the allocation

Let θ be the relative material angles and φ the angles of finite mechanical reservoirs, with velocities ω and ν. Positive diagonal matrices J, K and R_d describe driver inertia, torsional springs and relative friction. Eliminating the quasistatic Maxwell field produces an inductive inertia $I+L$. The reduced Lagrangian and reciprocal equations are

$$\begin{aligned} \mathcal{L}_{\mathrm{red}} &= \tfrac{1}{2} \boldsymbol{\omega}^T (I + L) \boldsymbol{\omega} + \tfrac{1}{2} \boldsymbol{\nu}^T J \boldsymbol{\nu} - \tfrac{1}{2} (\boldsymbol{\theta} - \boldsymbol{\phi})^T K (\boldsymbol{\theta} - \boldsymbol{\phi}), \\ (I + L) \dot{\boldsymbol{\omega}} &= -K(\boldsymbol{\theta} - \boldsymbol{\phi}) - R_d \boldsymbol{\omega}, \qquad J \dot{\boldsymbol{\nu}} = K(\boldsymbol{\theta} - \boldsymbol{\phi}). \end{aligned} \tag{75}$$

The induced electric field is already nonzero when a displaced spring first accelerates the current; its small storage is quantified below. The constraints maintain the spherical profiles and do no radial work. These springs and profiles specify a test material class, not a fluid planetary interior. Friction deposits heat in the same closed system. A resistive sheet realizes the torque coefficient through $R_{\mathrm{d,i}}=q_{\mathrm{i}}^{2}/(6\pi\Sigma_{\mathrm{i}})$, where Σ is sheet conductance. The complete reduced energy and the independent transfers obey

$$\begin{gathered} H = K_{\mathrm{rel}} + U_B + E_{\mathrm{driver}} + U_{\mathrm{heat}}, \qquad \dot{H} = 0, \\ E_{\mathrm{driver}} = \tfrac{1}{2}\boldsymbol{\nu}^T J \boldsymbol{\nu} + \tfrac{1}{2}(\boldsymbol{\theta}-\boldsymbol{\phi})^T K (\boldsymbol{\theta}-\boldsymbol{\phi}), \\ \dot{W}_{\mathrm{driver}} = -\boldsymbol{\omega}^T K(\boldsymbol{\theta}-\boldsymbol{\phi}), \quad \dot{E}_{\mathrm{driver}} = -\dot{W}_{\mathrm{driver}}, \quad \dot{U}_{\mathrm{heat}} = \boldsymbol{\omega}^T R_d \boldsymbol{\omega}, \\ \tfrac{d}{dt}(K_{\mathrm{rel}} + U_B) = \dot{W}_{\mathrm{driver}} - \dot{U}_{\mathrm{heat}}, \qquad \dot{U}_B = \boldsymbol{\omega}^T L \dot{\boldsymbol{\omega}}. \end{gathered} \tag{76}$$

In canonical variables, the current energy is one half of p^T(I+L)^{−1}p; adding magnetic energy again would double count it. The all-space field energy also already includes the exterior. At a finite enclosing boundary its transfer is checked directly from the induced electric field and the Poynting vector:

$$\oint_{r=R} \mathbf{S}\cdot d\mathbf{A} = \frac{\mu_0 m \dot{m}}{6\pi R^3} = \frac{dU_{\mathrm{out}}}{dt}, \qquad m = \sum_i m_i. \tag{77}$$

The one-shell lossless limit gives an analytic allocation before any numerical integration. Put A=I+L and start with a displaced driver, $\varphi=\varphi_0$, with zero velocities and current. The initial energy is $E_0=k\varphi_0^{\,2}/2$. Then

$$\begin{gathered} \Omega^2 = k\left(\tfrac{1}{A}+\tfrac{1}{J}\right), \qquad \frac{U_B(t)}{E_0} = \frac{L}{A}\frac{J}{A+J}\sin^2(\Omega t), \\ \frac{U_{\mathrm{out}}(t)}{E_0} = \frac{1}{3}\frac{U_B(t)}{E_0}. \end{gathered} \tag{78}$$

The factor L/A is the inductive share of the current-mode energy; $J/(A+J)$ accounts for recoil of the finite driver. Both follow from the supplied material and geometry. The two-shell test adds mutual induction, signed interference and returned heat to this analytic limit.

Choose radius, time and inertia units a_0, t_0, I_0, charge unit [18πI_0/(μ$_0$$a_0$)]^{1/2}, and energy unit $I_0/t_0^{\,2}$. The slow-time scale ct_0/a_0=10000 is specified before evolving. The remaining dimensionless inputs are all new test data:

Input	Inner	Outer
Radius a; charge q	0.55; 1	1; 1
Relative inertia I; driver inertia J	0.20; 0.70	0.40; 1.10
Spring K; resistance Rd	0.80; 0.03	1.20; 0.05
Initial driver angle: aligned / opposed	0.20 / 0.20	0.20 / −0.20

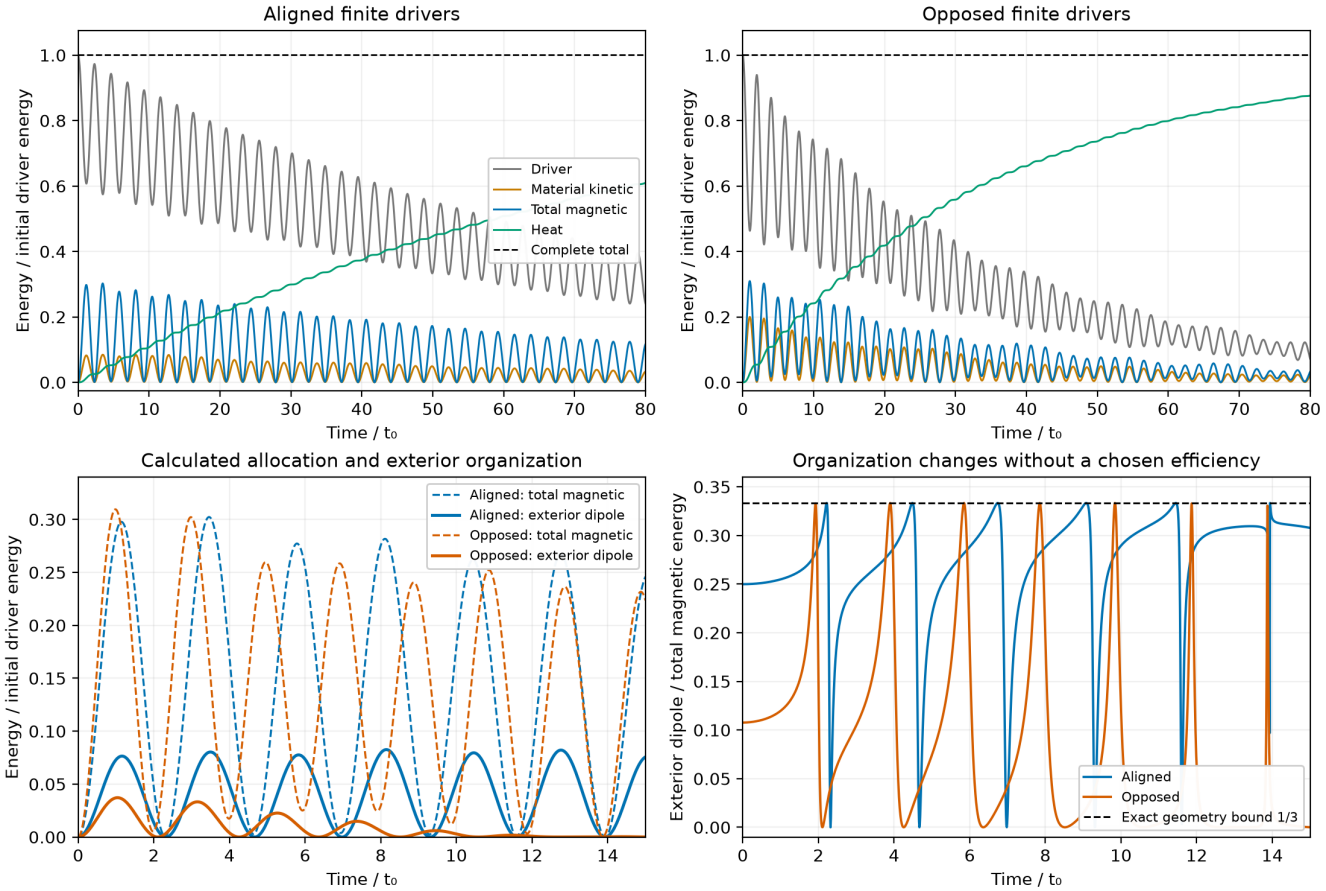
Both preparations have zero initial current, magnetic field and velocities, and the same available energy E_0=0.04. Only the relative signs of the initial driver displacements differ. The inductive

matrix has entries L_{11}=0.55, L_{22}=1 and L_{12}=L_{21}=0.3025. Its eigenvalues are 0.397997 and 1.152003; those of I+L are 0.631005 and 1.518995. The material and field quadratic energies are therefore positive.

25. Calculated organization, convergence and limits

Preparation	Peak UB / E0	Peak Uout / E0	Exterior peak time / t0
Aligned	30.2279%	8.2558%	8.147262
Opposed	30.9534%	3.7161%	1.040644

These are computed peaks over 80 time units; total and exterior maxima need not occur together. Nearly equal maximum total magnetic allocation gives a 2.22-fold difference in peak exterior dipole energy. The calculation supplies the transfer and organization instead of installing either as an efficiency. It is a finite conversion and decay experiment, not a self-sustaining dynamo or a selected long-term amplitude.



Finite-driver allocation from the coupled equations. The complete total includes material kinetic energy, driver energy, total magnetic energy and heat. Exterior dipole energy is part of the total magnetic energy, not an additional term. The lower panels show the early evolution and the exact organization bound. Both preparations begin with the same finite driver energy and zero current.

Material inertia strengthens the energy allowance. With $A=I+L$ and $d_{\mathrm{i}}=q_{\mathrm{i}}a_{\mathrm{i}}^{2}$ in the specified units, the Rayleigh bound is

$$\frac{U_{\mathrm{out}}}{K_{\mathrm{rel}}+U_{B}}\le\frac{\mathbf{d}^{T}A^{-1}\mathbf{d}}{3}=0.241732145,\qquad\frac{U_{\mathrm{out}}}{E_{0}}\le0.241732145.\tag{79}$$

This bound follows from the declared inertias and geometry; neither preparation saturates it. A fixed volume and the same complete reduced energy therefore permit different exterior histories. The source conversion is consistent with both at this approximation order; their omitted electric and radiative corrections and a full nonspherical source boundary have not been equated. Its numerical value does not choose one without the material class and preparation.

Checks and controlled approximation

Time steps 0.08, 0.04, 0.02 and 0.01 show fourth-order state convergence against an independent matrix-exponential solution. At the finest step the state errors are below 2.3×10^{−9} and the largest relative complete-energy error is 4.65×10^{−10}. Four power ledgers are integrated separately, including the outer Poynting flux; their absolute residuals are below 3×10^{−11}. Regional field integrals independently reproduce the inductive quadratic form. A deliberately inconsistent half-strength induction feedback produces a 31.55% energy error and is rejected. A lossless control retains conservative exchange with zero heat.

The maximum natural frequency satisfies $\omega_{\max}a_{0}/c=1.5844\times10^{-4}$, and all relative surface speeds satisfy $v/c<1.506\times10^{-5}$. Evaluating the leading omitted induction electric storage gives at most 1.93×10^{-9} of the initial energy. The leading magnetic-dipole radiation estimate over the full interval is below 1.52×10^{-11} of that energy; the conservative maximum-power-times-duration estimate is below 4.46×10^{-11}. Generic retardation corrections scale as $(\omega a/c)^{2}$. These are consistency diagnostics of the reduction, not a claim of an exact radiative Maxwell solution or a rigorous bound on every observable.

Reproduction and the full derivation are in *evolution/FINITE-DRIVER-AND-ORGANIZATION.txt*, *evolve_finite_inductive_driver.py*, *check_exact_nested_energy.py* and their JSON audit records. The stronger conclusion is limited and constructive: a declared charged-material model can calculate the allocation, while a source-energy scalar alone still does not select the observed planetary magnetic field.

26. An additional capacity cross-term and its numerical test

A further proposed route is to retain the entire clock departure from the unloaded reference, rather than only the difference between held and rotating clocks. This gives another exact identity on the same positive-clock branch:

$$1-\beta=(1-\alpha)+(\alpha-\beta),$$
$$Mc^2(1-\beta)^2=U_{GG}+U_\times+U_*,$$
$$U_{GG}=Mc^2(1-\alpha)^2,\qquad U_\times=2Mc^2(1-\alpha)(\alpha-\beta). \tag{80}$$

All three terms are nonnegative for $0\le\beta\le\alpha\le1$. The identity is worth keeping because it separates held burden, rotational clock change and their product. Naming those terms does not by itself establish separate physical reservoirs or the work that can be extracted from them.

$$U_\times\simeq\frac{MS^2v^2}{2c^2},\qquad U_*\simeq\frac{Mv^4}{4c^2},\qquad \frac{U_\times}{U_*}=\frac{2(1-\alpha)}{\alpha-\beta},$$
$$b_\times=c\sqrt{2\mu_0\bar{\rho}(1-\alpha)(\alpha-\beta)}\,. \tag{81}$$

The last expression tests the additional assignment U_d,out=U_×. It is evaluated with the same masses, radii, rotations, measured dipoles and solar epochs as before. The exact rationalized clock differences are used; no efficiency is chosen from the observations.

Body	Cross-term candidate	Candidate / observed
Mercury	2.50458 μT	13.17
Earth	1020.28 μT	34.31
Jupiter	71,264.08 μT	159.56
Saturn	23,858.61 μT	1018.03
Sun	1222.37 G	485-1397 across five minima

Mercury’s measured exterior dipole energy is about 0.576% of U_×. Thus the cross-term is numerically large enough to avoid the earlier U*-only budget objection if an actual source process makes it available. It does not select that fraction. Its unchanged direct magnetic assignment overpredicts every listed object. Keeping U* as well gives b=√(b_×²+b₀²), which raises every prediction further; retaining the entire squared deficit raises it again. These outcomes preserve a potentially useful mixed capacity scale while separating it from a successful magnetic allocation law.

Two exact distinctions prevent false energy matches

The retained shell-selection energy W_req and this squared clock quantity have different leading orders. If their common source-energy label is E_s=Mc² and u=1−α²,

$$W_{\mathrm{req}} = Mc^2\frac{1-\alpha}{1+\alpha}, \qquad U_{GG} = uW_{\mathrm{req}},$$
$$W_{\mathrm{req}} \simeq \frac{Mc^2u}{4}, \qquad U_{GG} \simeq \frac{Mc^2u^2}{4}. \tag{82}$$

U_GG is therefore not the leading approximation to W_req. Similarly, the exterior energy must be integrated over its radius-dependent volume. For one unchanged dipole generated inside a,

$$b(R) = b(a)\left(\frac{a}{R}\right)^3,$$
$$\frac{b(R)^2}{b(a)^2} = \left(\frac{a}{R}\right)^6, \qquad \frac{U_{\mathrm{out}}(R)}{U_{\mathrm{out}}(a)} = \left(\frac{a}{R}\right)^3. \tag{83}$$

The sixth power applies to the squared field or its local energy density. The integrated exterior energy has a third power. Conducting screening is a further dynamical calculation involving conductivity, frequency and geometry; it cannot be inserted as an additional universal depth factor.

The proposed material story also has to retain the pressure and collision equations. Universal gravitational acceleration cancels from the relative acceleration of freely falling species. A supported plasma can polarize because its pressures and supports act differently, but the coefficient of its electric field follows from that complete balance. Common neutral co-rotation cancels its transport current; it does not prohibit independent conduction currents or magnetization in a solid. No Rossby-number saturation relation has been derived here that explains the Earth or solar coincidences.

27. Reconciling the phase connection, clock weighting and charged exterior

A local charged phase connection is compatible with the retained probe coupling. Once the charged phase, its local symmetry and its potentials are admitted, the probe action is

$$L_p = -E_0\beta - e\Phi + eA_iv^i, \qquad \beta^2 = \alpha^2 - q_{ij}v^iv^j/c^2,$$
$$P_i = \frac{mq_{ij}v^j}{\beta} + eA_i, \qquad H_p = \alpha\sqrt{E_0^2 + c^2q^{ij}(P_i - eA_i)(P_j - eA_j)} + e\Phi. \tag{84}$$

Its motion equation contains both the burden gradient and the Lorentz response. On a zero-shift background with constant rest mass and charge,

$$\frac{d}{dT}\left(\frac{mq_{ij}v^j}{\beta}\right) - \frac{m}{2\beta}(\partial_iq_{jk})v^jv^k$$
$$= \frac{m}{2\beta}\partial_iS^2 + e(E_i + \sqrt{q}\,\varepsilon_{ijk}v^jB^k). \tag{85}$$

Here ε denotes the alternating symbol, q=det(qᵢⱼ), E_i=−∂ᵢΦ−∂_TA_i and Bⁱ=εⁱʲᵏ∂ⱼAₖ/√q. The two homogeneous Maxwell identities follow locally from these definitions. A global potential and absence of magnetic monopoles require the corresponding topological domain. Local phase invariance motivates this connection after the charged sector is supplied; the neutral capacity scalar does not alone establish that sector or its numerical charge.

The mode-clock rule weights a supplied local electromagnetic energy by the lapse. With the retained Maxwell quadratic constitutive energy, the corresponding field Lagrangian is

$$L_{\mathrm{EM}} = \int \left[\frac{\sqrt{q}}{2} \left(\frac{\varepsilon_0}{\alpha} q^{ij} E_i E_j - \frac{\alpha}{\mu_0} q_{ij} B^i B^j \right) - \rho_c \Phi + J_c^i A_i \right] d^3x. \tag{86}$$

The sources here are coordinate densities: ρ_c=√q ρ_held and J_cⁱ=α√q j_heldⁱ for zero shift. Varying the potentials gives the sourced equations. This is a useful consistency derivation of Maxwell propagation in the supplied capacity geometry. The choice of Maxwell's quadratic local energy is selected only inside the field class and propagation conditions examined earlier; lapse weighting alone does not establish it.

$$\begin{aligned} \partial_i \left(\frac{\varepsilon_0 \sqrt{q}}{\alpha} q^{ij} E_j \right) &= \rho_c, \\ \varepsilon^{ijk} \partial_j \left(\frac{\alpha}{\mu_0} q_{kl} B^l \right) - \partial_T \left(\frac{\varepsilon_0 \sqrt{q}}{\alpha} q^{ij} E_j \right) &= J_c^i. \end{aligned} \tag{87}$$

In an isotropic static ruler qᵢⱼ=ψ⁴δᵢⱼ, the coordinate optical coefficients are ε_eff=ε₀ψ²/α and μ_eff=μ₀ψ²/α. Thus coordinate speed is cα/ψ², physical distance per reference time is cα, and the held local clock measures c. Their ratio preserves Z₀. Constant impedance alone does not eliminate all reflection or scattering: geometry, angular propagation and effective wave potentials remain. Even equal-impedance flat interfaces can reflect obliquely. The wave-potential example is independently discussed by Crispino, Dolan and Oliveira (2009).

The charged burden and lapse have matching coefficients

In the selected static spherical charged branch, let γ_q=Gq_e²/(4πε₀c⁴), with total source charge q_e, A=α² and B_rad=A⁻¹. The radial source and stress equations give

$$\begin{aligned} (r^2 A')' &= \frac{2\gamma_q}{r^2}, \qquad A = 1 - \frac{2GM}{c^2 r} + \frac{\gamma_q}{r^2}, \\ S^2 &= c^2(1 - A) = \frac{2GM}{r} - \frac{G q_e^2}{4\pi\varepsilon_0 c^2 r^2}. \end{aligned} \tag{88}$$

The charge coefficient in S² must be 1/(4π), matching A; doubling it would violate S²=c²(1−A). Electromagnetic spacetime stress has vanishing four-dimensional trace, whereas its spatial stress trace in this convention equals its positive energy density. Both the radial ruler equation and the lapse/work equation are used here. This recovers the retained charged exterior consistently; it does not obtain B_rad=A⁻¹ from lapse weighting alone.

The adjacent exploration therefore contributes an exact cross identity, a numerical discrimination and a clean reconciliation of known charged probe and field equations. The allocation results developed earlier remain the direct answer to how a specified source transfers energy into an organized magnetic field.

28. What determines the organized magnetic energy

The source law keeps the complete accounting in view. The common energy supplies the material and field equations together. Once signed carrier state, geometry, material response and preparation are specified, their evolution determines the magnetic allocation. This has been carried out explicitly for regular current modes and finite drivers.

For one lossless neutral shell and a finite driver, the result takes a particularly compact form. Let I be the carrier relative inertia, L its calculated inductive inertia, J the driver inertia and k the torsional stiffness. Starting with E_0 stored in the torsional spring, both material and driver velocities zero and no magnetic field,

$$\frac{U_B(t)}{E_0} = \frac{L}{I+L}\frac{J}{I+L+J}\sin^2(\Omega t),$$
$$\Omega^2 = k\left(\frac{1}{I+L} + \frac{1}{J}\right), \qquad U_{d,\,\mathrm{out}} = \frac{U_B}{3}. \tag{89}$$

The first fraction is the field share of the current response. The second is the finite-driver recoil. The last third is the exterior share selected by this shell geometry. All three have separate derivations. Uniform-density rigid relative rotation in a regular volume gives 7/30 in place of the shell's one third. Nested and distributed currents retain their signed mutual terms.

This answers the allocation question within a specified material class. It also identifies precisely why one number for total source energy cannot replace a physical source state. In the two-shell test, the same reduced initial energy gives peak exterior shares 8.2558% and 3.7161% over $0 \le t/t_0 \le 80$ under different signed preparations. Both follow the unchanged coupled equations. At fixed reference volume, a complete source can conserve Q_s while its current organization evolves.

The spatial variational result adds a complementary tool. It determines the smallest carrier-plus-field cost of a prescribed dipole and therefore the largest possible dipole for a verified current-sector budget. The budget, its driving mechanism and the attained state are separate quantities. A bound does not establish saturation, and a stationary minimum does not establish a dissipative attractor.

The next material calculation is now specific

For Earth, the retained burden profile supplies composition work, the matched thermal state supplies phase evolution and entropy allowances, and the electromagnetic equations supply induction, reciprocal force and heat return. The remaining terrestrial calculation is the evolution of those independently constrained material data into the signed current and its exterior dipole in the relevant parameter regime. The benchmark calculations validate components of that route; they do not yet supply that complete planetary amplitude.

The largest theoretical gain is therefore a usable bridge: source energy can be converted into an organized magnetic field by one common generator, with the conversion, geometry and feedback calculated separately and then checked together. The scalar invariant and source law participate throughout. The charged representation and material state remain explicit parts of the electromagnetic realization, rather than being hidden inside a fitted magnetic fraction.

29. Verification, retention and reproducibility

The publication baseline is identified by its immutable Git commit in [R1]. The fourteen subsequent research records are indexed in the companion archive. Existing results are retained with their accepted numerical limits. The archive separates those carry-forward calculations from the new current-organization and finite-driver experiments.

New calculation	Independent checks
Uniform neutral ball	Ten symbolic identities; direct field integration, current-potential energy and inertia partition.
Constrained current and geometry	Thirty-three exact checks; six direct quadratures with maximum residual 8.95×10^{-11}; regularity, matching and asymptotic limits.
Finite inductive driver	Fourteen symbolic checks; regional field integrals; four time-step refinements against a matrix exponential; separate work, heat and boundary-flux integrals.
Terrestrial dipole energy history	Same-radius conversion of the retained IGRF14 coefficients; independent squared-amplitude energy conversion; input hash retained.
Additional capacity cross-term	Forty-nine high-precision checks; unchanged planetary and solar inputs; exact clock remainder and radius-scaling identities.
Cross-review	Separate reviews of signs, constrained variation, source-versus-screen readings, units, comparison observables and approximation scope.

Numerical precision is reported as a reproducibility check, not as observational accuracy. Model alternatives and empirical uncertainties are not replaced by integration error. No high-precision residual is presented as a probability that SET is correct. The paper preserves the positive results and the control cases needed to decide where the next physical calculation belongs.

30. References and reproducibility

The labels distinguish the published SET baseline [R1], the subsequent local research records [N1-N14], independent material sources [M1-M5], and observational or numerical input sources [D1-D12]. The local SET records contain derivations and computational controls; they have not been independently peer reviewed. External references supply the stated measurements, material models or standard theoretical comparisons. Their use does not independently validate the SET-specific deductions.

Published baseline and subsequent SET records

[R1] Space Emanation Theory research (2026). *Source, clock and interacting-field energy.* Research synthesis published 6 November 2026, 44 pages. PDF at publication commit 79e3489.

The following records began on 6 November 2026; some final checks completed after local midnight. Their exact folder and file mappings are supplied in the companion archive’s provenance map. Titles below identify the research content rather than implying separate journal publication.

[N1] *Vacuum impedance, source selection and Maxwell normalization.* Constant field calibration, independent source-boundary tests and normalization freedom.

[N2] *Charged matter: capacity, common energy and source selection.* Coupling-square selection, charged work and surviving supported-source coupling families.

[N3] *Charge-state origin, phase-current evolution and neutral controls.* Polar canonical transport, signed rearrangement, phase-energy bounds and elementary quantum controls.

[N4] *Capacity, signed internal motion and electromagnetic coupling.* Energy sums, signed generators, composite-connection tests and observational phase constraints.

[N5] *Internal symmetry and electric charge.* Observation-conditioned internal symmetry, Gauss boundary charge and neutral counterexamples.

[N6] *Dynamical electromagnetic source closure.* Explicit source functional, evolving drivers, complete energy shares, photon work and ordered quantum clocks.

[N7] *Earth magnetic naive tests.* Initial dimensional and rotating-charge trials, with all comparison choices retained.

[N8] *Earth signed-current evolution.* Two-fluid current dynamics, spatial dipole completion and energy/dissipation identities.

[N9] *Independent core material, gravitational work and Earth's magnetic field.* PREM burden, compositional work, prescribed-flux budgets and conducting-core decay.

[N10] *Matched core-mantle thermal source and coupled magnetic controls.* Consistent interface, phase branches, finite source feedback and bounded global solver controls.

[N11] *Rotational clock amplitude and Earth's magnetic scale.* Exact clock remainder, known-target comparison, spatial alternatives and current completion.

[N12] *Boundary clock energy, source selection and a Jupiter transfer test.* Relative-action selector, exterior work flux and the unchanged second-object test.

[N13] *Magnetic corrections: inputs, algebra, organization and exterior energy.* Twenty declared alternatives, solar rotation projection and multipole energy audit.

[N14] *Magnetic fields: measurement history and source organization.* Historical inversions, physical epoch changes, independent active domains, Mars and Venus controls.

Independent material inputs and numerical benchmarks

[M1] Dziewonski, A. M., and Anderson, D. L. (1981). *Preliminary reference Earth model.* Physics of the Earth and Planetary Interiors. doi:10.1016/0031-9201(81)90046-7. Density, seismic speeds and reference geometry used in the radial Earth calculation.

[M2] Pozzo, M., Davies, C., Gubbins, D., and Alfè, D. (2013). *Transport properties for liquid silicon-oxygen-iron mixtures at Earth's core conditions.* Physical Review B 87, 014110. doi:10.1103/PhysRevB.87.014110. Independent liquid-alloy electrical and thermal transport.

[M3] Gubbins, D., Alfè, D., Davies, C., and Pozzo, M. (2015). *On core convection and the geodynamo: Effects of high electrical and thermal conductivity.* Physics of the Earth and Planetary Interiors 247, 56-64. doi:10.1016/j.pepi.2015.04.002; author manuscript. Paired core material and energy coefficients.

[M4] Frost, D. A., et al. (2022). *Multidisciplinary constraints on the thermal-chemical boundary between Earth's core and mantle.* Geochemistry, Geophysics, Geosystems 23, e2021GC009764. doi:10.1029/2021GC009764. The present calculation uses the declared mantle boundary-layer response, not the paper's geomagnetic constraint as a magnetic normalization.

[M5] Christensen, U. R., et al. (2001). *A numerical dynamo benchmark.* Physics of the Earth and Planetary Interiors 128, 25-34. doi:10.1016/S0031-9201(01)00275-8. Solver verification reference. The MAG and MagIC repositories provide the associated implementations and further test provenance.

Additional material branches use Pozzo et al. (2014), solid-core conductivity; Dannberg et al. (2024), mantle heat-flow modeling and its data archive; and Wong et al. (2018), liquidus material model. Aubert (2026) supplies an Earth-regime feasibility comparison whose forcing uses geomagnetic-variation information; it is not imported as the present independent field prediction.

Bulk inputs, magnetic observations and constants

[D1] NASA/JPL Solar System Dynamics. Planetary physical parameters; astrodynamic parameters. Mass, mean radius and sidereal period are frozen in the comparison scripts. Solar parameters and the declared differential-rotation profile are linked through the NASA Sun Fact Sheet.

[D2] NOAA/NCEI and British Geological Survey. World Magnetic Model 2025; WMM2025 coefficient archive. Internal harmonics supply the original Earth dipole and the separate degree-12 exterior-energy comparison.

[D3] IAGA/NCEI. International Geomagnetic Reference Field, generation 14; official coefficients. The 1900-2025 epochs describe physical field evolution in one modern reconstruction.

[D4] Connerney, J. E. P., et al. (2022). *A new model of Jupiter's magnetic field at the completion of Juno's prime mission.* Journal of Geophysical Research: Planets 127, e2021JE007055. doi:10.1029/2021JE007055; NASA PDS JRM33 coefficients.

[D5] Cao, H., et al. (2020). *The landscape of Saturn's internal magnetic field from the Cassini Grand Finale.* Icarus 344, 113541. doi:10.1016/j.icarus.2019.113541; accepted manuscript. This supplies Cassini11+. The earlier Cassini11 multipoles use Dougherty et al. (2018) and the NASA PDS table. The history audit also uses Moore et al. (2021).

[D6] Philpott, L. C., et al. (2014). *Constraints on the secular variation of Mercury's magnetic field from the combined analysis of MESSENGER and Mariner 10 data*. Geophysical Research Letters 41, 6627-6634. doi:10.1002/2014GL061401. The original orbital comparison separately uses Anderson et al. (2012).

[D7] Wardinski, I., Amit, H., Langlais, B., and Thébault, E. (2021). *The internal structure of Mercury's core inferred from magnetic observations.* Journal of Geophysical Research: Planets 126, e2020JE006792. doi:10.1029/2020JE006792. Full-mission inversions retain the spread from distinct regularization priors.

[D8] Wang, Y.-M. (2024). *The radial interplanetary field strength at sunspot minimum as polar field proxy and solar cycle predictor.* Astrophysical Journal Letters. doi:10.3847/2041-8213/ad1c65. Photospheric axial coefficients are compared through |D_ax|/2. Calibration and instrument comparisons additionally use Wang, Ulrich and Harvey (2022) and Upton and Hathaway (2023).

[D9] Connerney, J. E. P., et al. (2001). *The global magnetic field of Mars and implications for crustal evolution.* Geophysical Research Letters 28, 4015-4018. NASA author/mission article. Its global equatorial dipole limit is distinct from local remanent crustal fields.

[D10] Phillips, J. L., and Russell, C. T. (1987). *Upper limit on the intrinsic magnetic field of Venus.* Journal of Geophysical Research 92, 2253-2263. doi:10.1029/JA092iA03p02253. The quoted moment parameter has units T m^3, with its electromagnetic conversion already included.

[D11] Preising, M., et al. (2023). *Material properties of Saturn's interior from ab initio simulations.* Astrophysical Journal Supplement Series 269, 47. doi:10.3847/1538-4365/ad0293. Density and transport tables supply the declared metallic-shell trial; its boundaries are a material-model result.

[D12] NIST/CODATA. *Recommended values of the fundamental physical constants: 2022*. NIST SP 961, May 2024. constants table; machine-readable values. The scripts retain the adopted values and uncertainty conventions.

Phase interpretation and the research archive

The charged-state interpretation is constrained by Tonomura et al. (1986), shielded-flux interference; NIST Josephson standards; Higgs (1964), the charged-condensate mass mechanism; and Gralla and Jacobson (2014), Euler-potential field descriptions. The hydrogen check uses the NIST ionization datum. These identify supplied physics and tests of interpretation.

The companion SET-source-energy-magnetic-organization-research.zip is the reproducibility entry point. Its inventory identifies the synthesis source, research records, calculation scripts, retained numerical exports and external dependencies included for this publication. It separates new calculations from imported controls and records accepted and rejected numerical cases.

PROVENANCE-MAP.txt maps [R1] and [N1-N14] to their detailed records. *RETENTION-DEPENDENCIES.txt* classifies identities, conditional selections, numerical tests and superseded claims. *NUMERICAL-CARRYFORWARD.json* preserves twelve existing numerical exports with their source hashes; its collection script copies those results without fitting or reinterpreting them. The build source supplies numbered equations and the figure inputs; the validation record distinguishes mathematical parsing from final visual review.

For an empirical comparison, reproduce the frozen bulk inputs first, then the stated harmonic norm or energy sum at the common radius. For a dynamical control, reproduce its declared material law, initial and boundary data, resolution and conservation checks before interpreting its output. Roundoff precision, numerical convergence, material uncertainty and observational model variation remain distinct quantities throughout the archive.